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(54) **HIGH-STRENGTH ALUMINUM ALLOYS
FOR CAN END STOCK AND METHODS FOR
PREPARING THE SAME**

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(57) **ABSTRACT**

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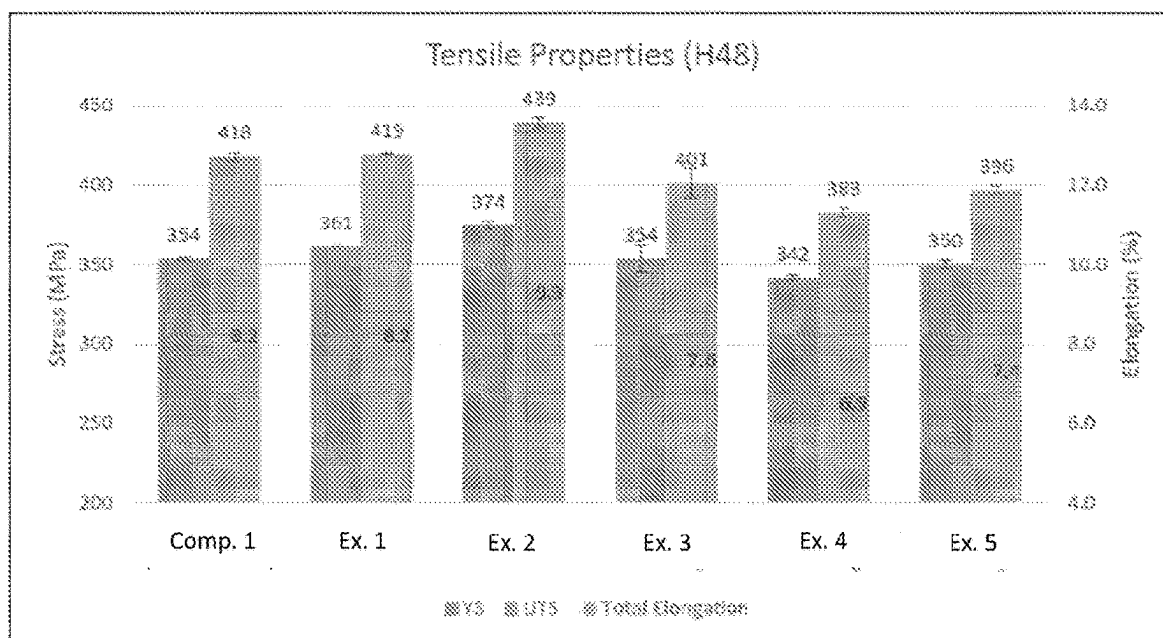
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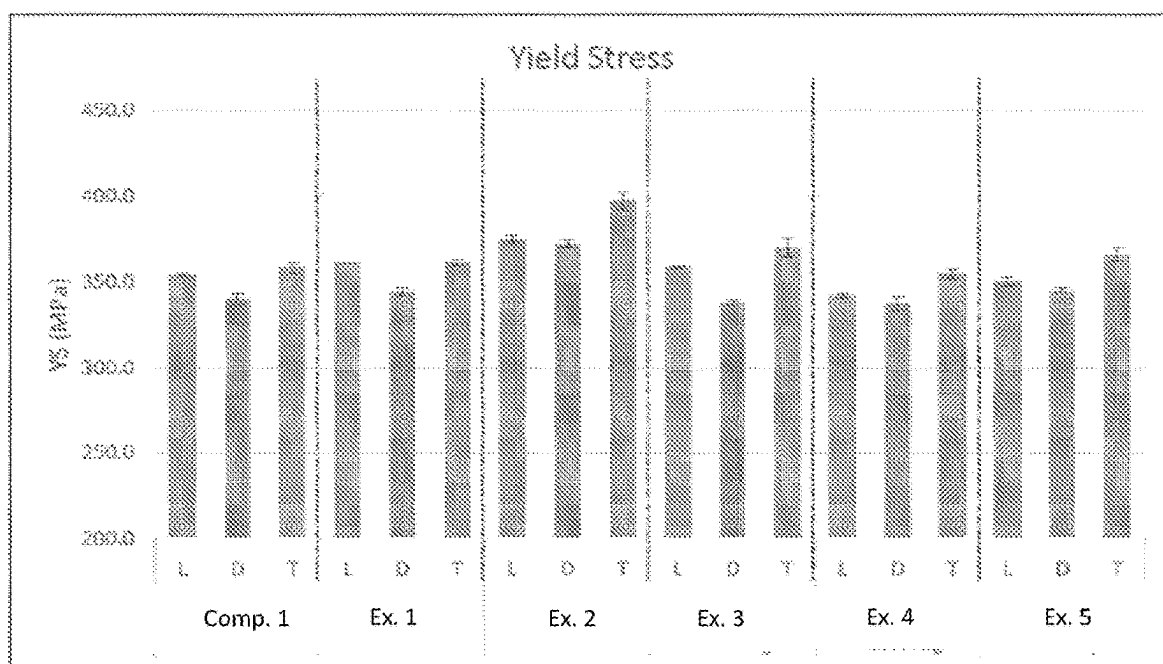
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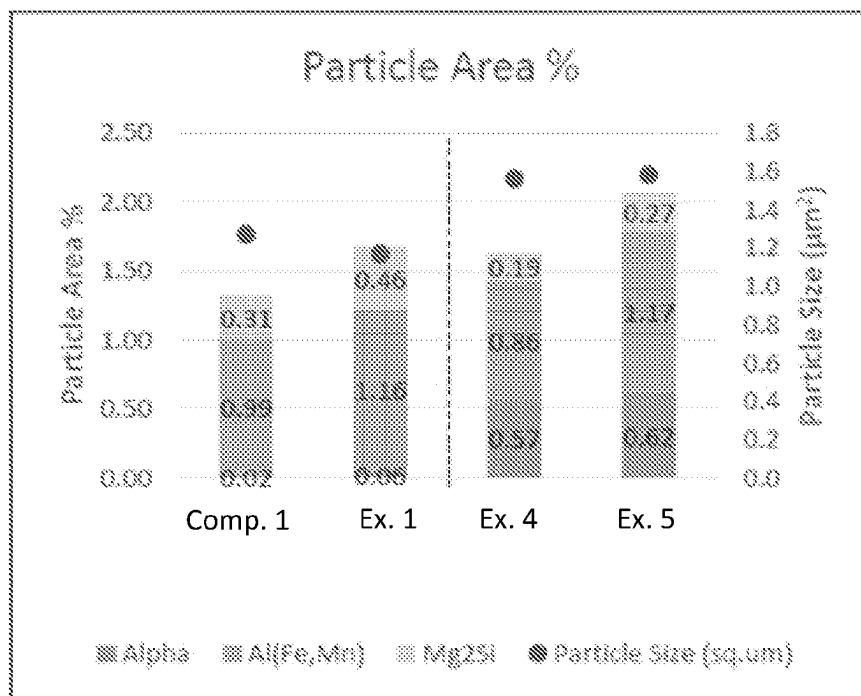
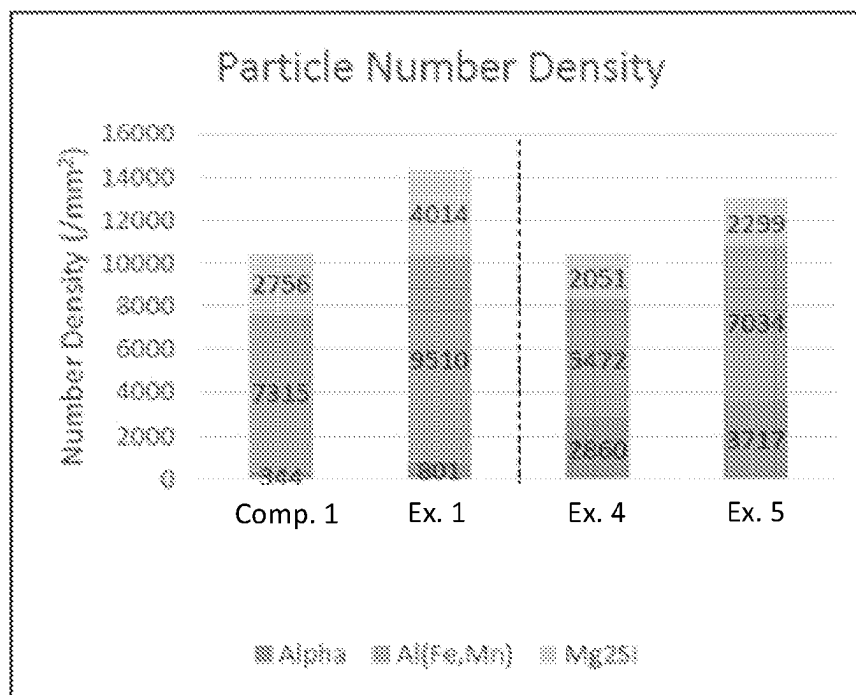
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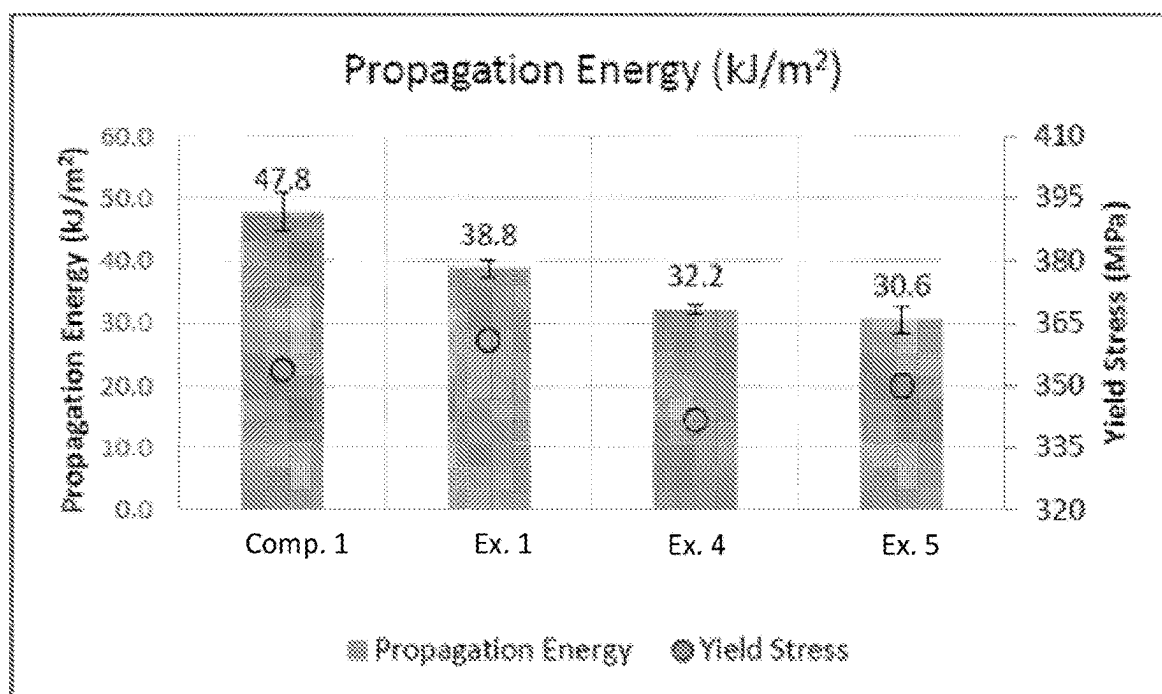
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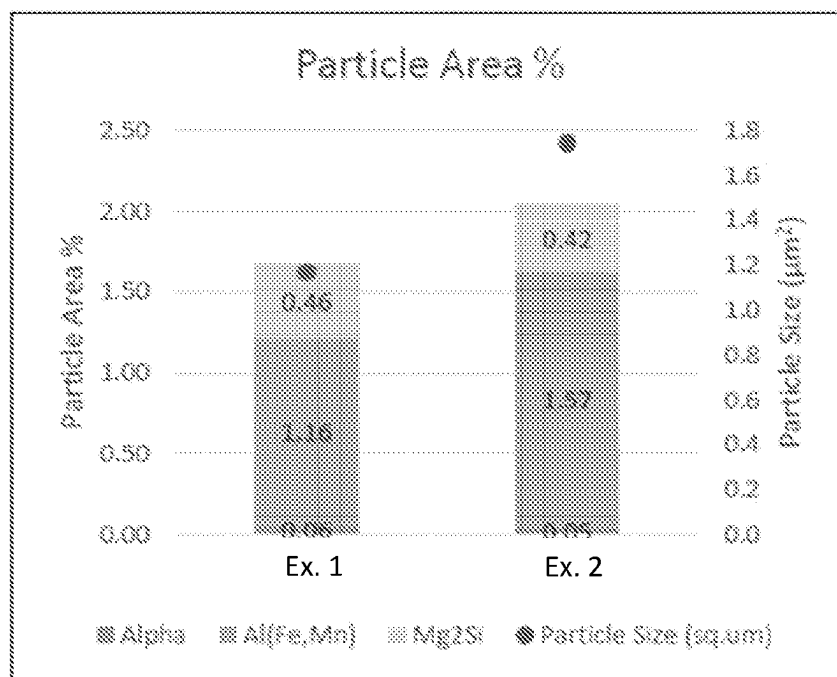
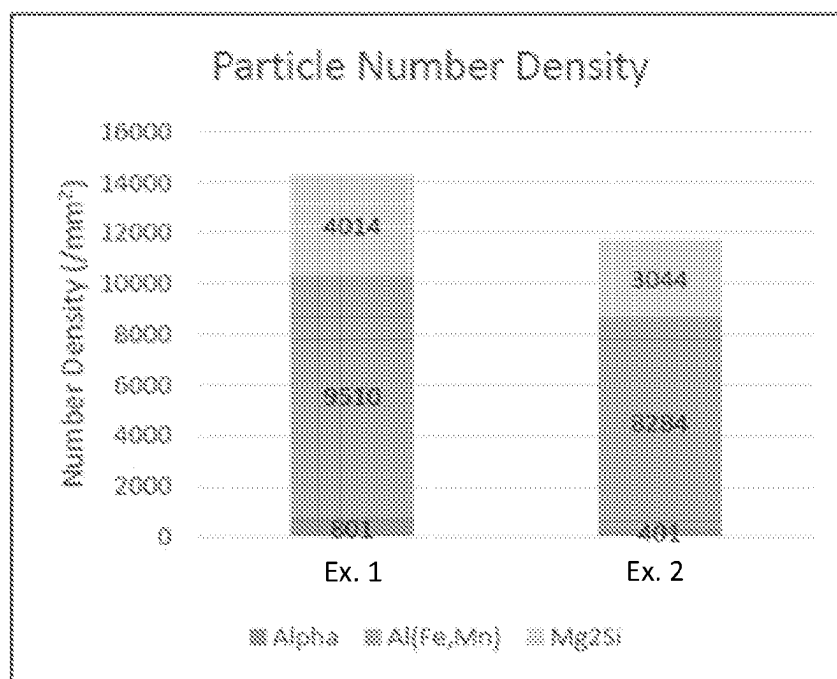
Described herein are novel aluminum alloys including recycled aluminum alloy materials which exhibit high strength and high formability. The aluminum alloys described herein, which are suitable for use as can end stock, for example, exhibit high strength and formability despite having a lower Mg content than traditional AA5182 aluminum alloys used to produce can end stock. The present disclosure provides a cost-effective alternative to the use of AA5182 alloy for can end stock.

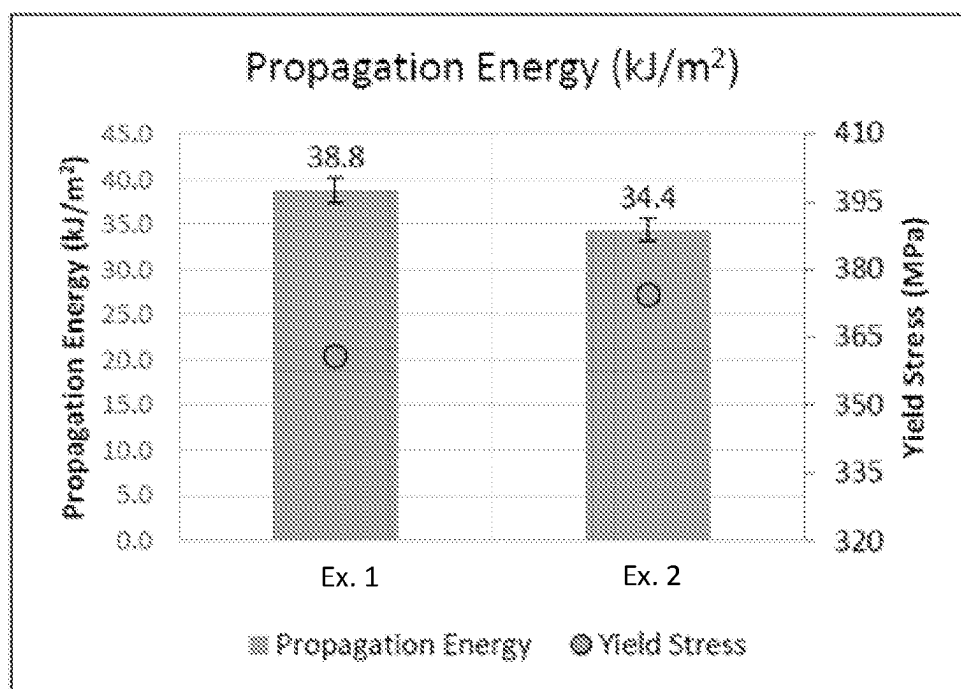
**FIG. 1**

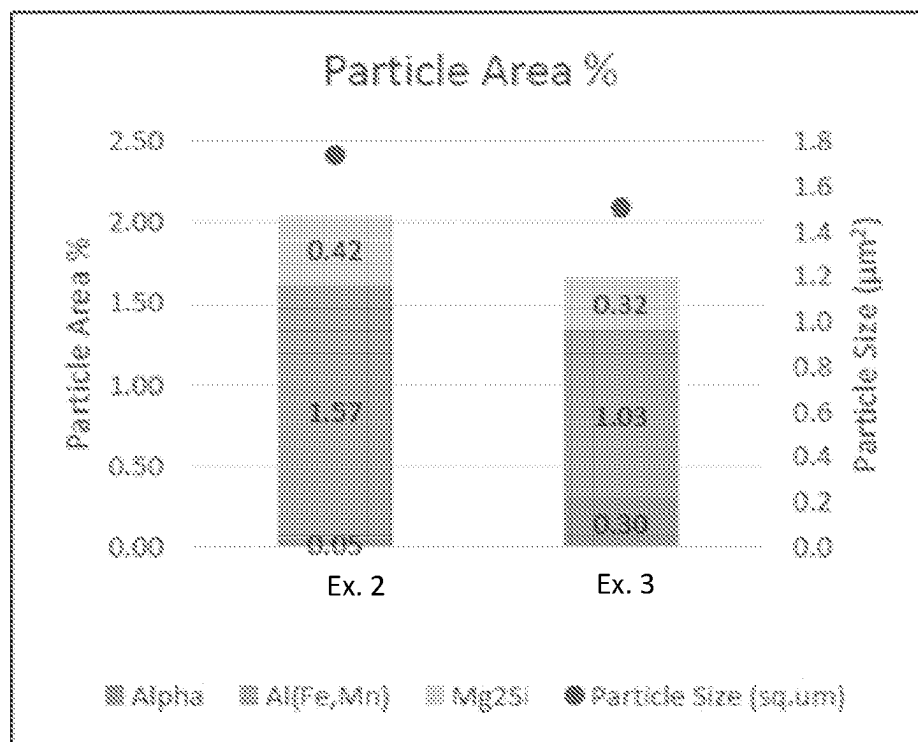
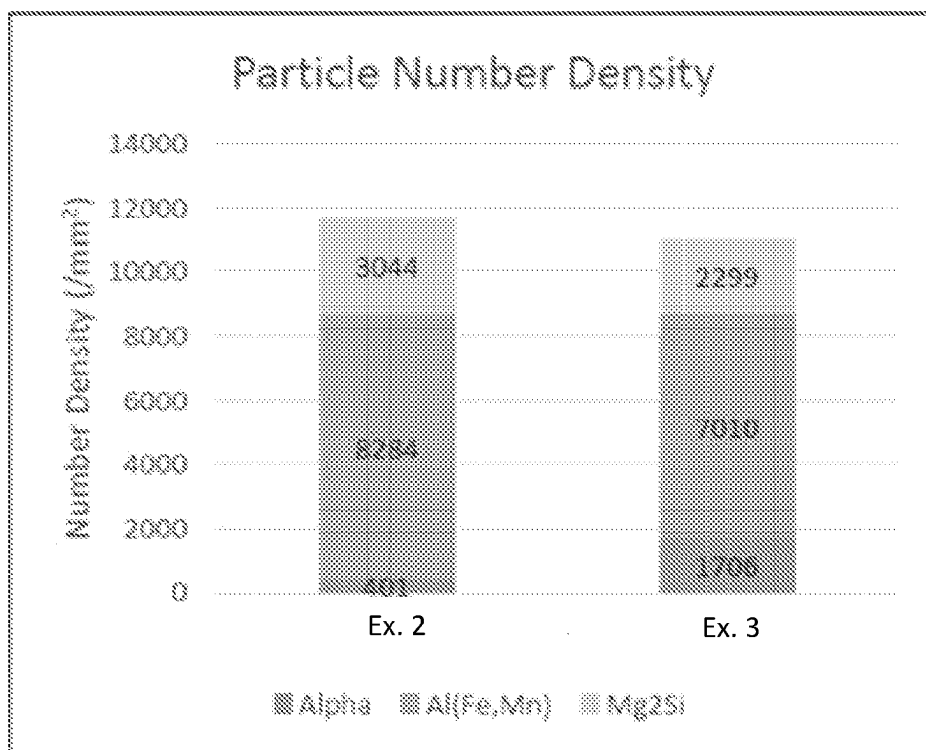
**FIG. 2**

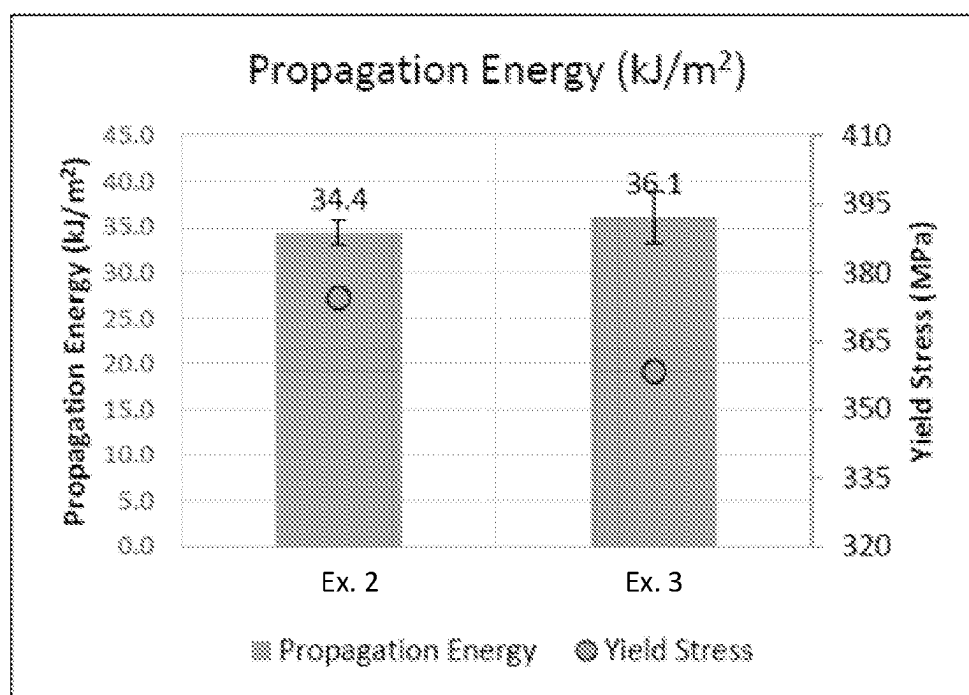
**FIG. 3A****FIG. 3B**

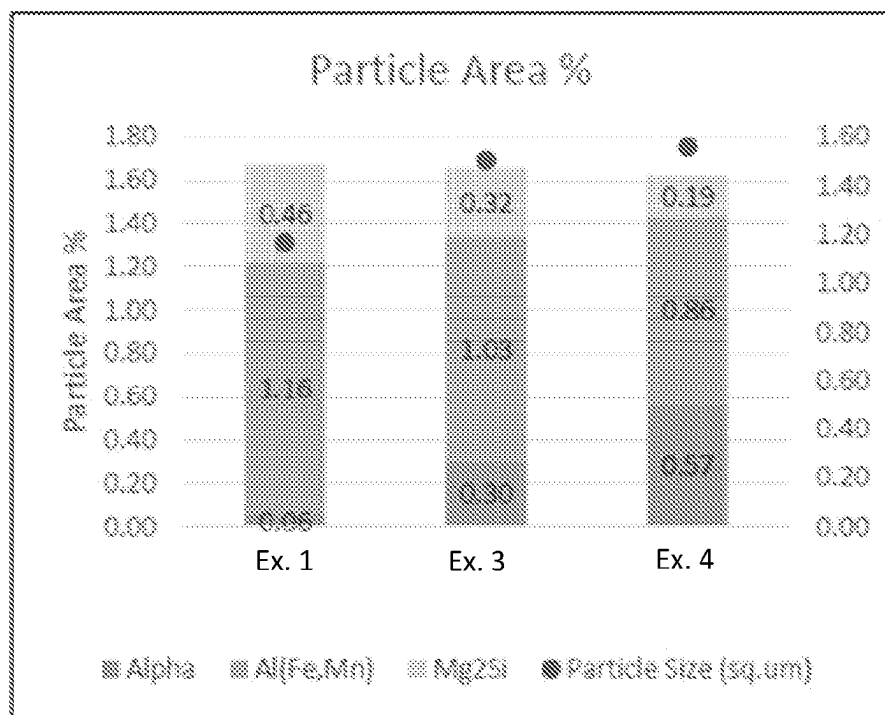
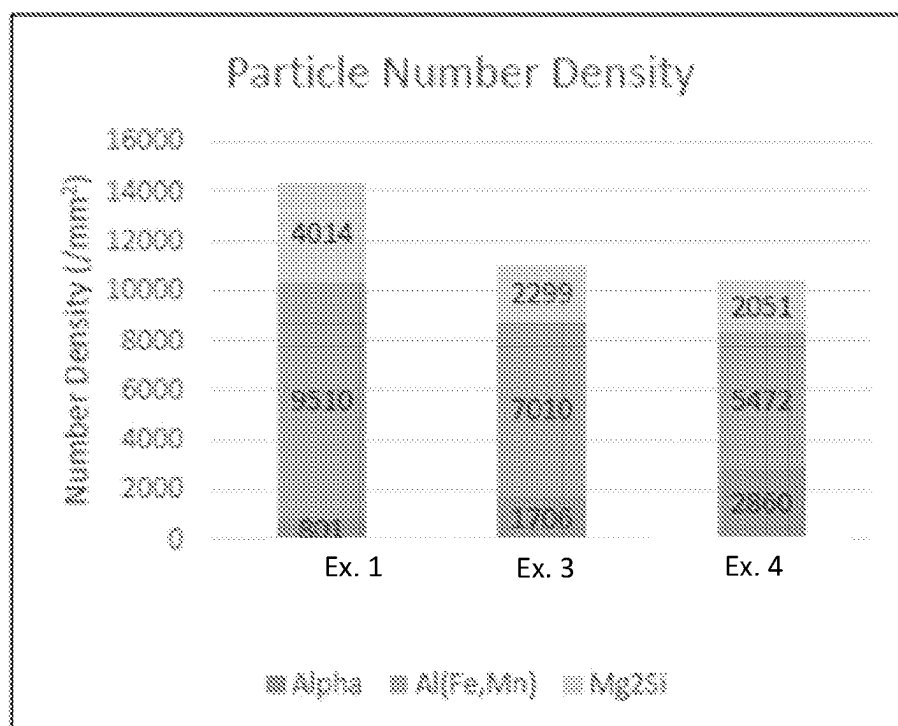
**FIG. 4**

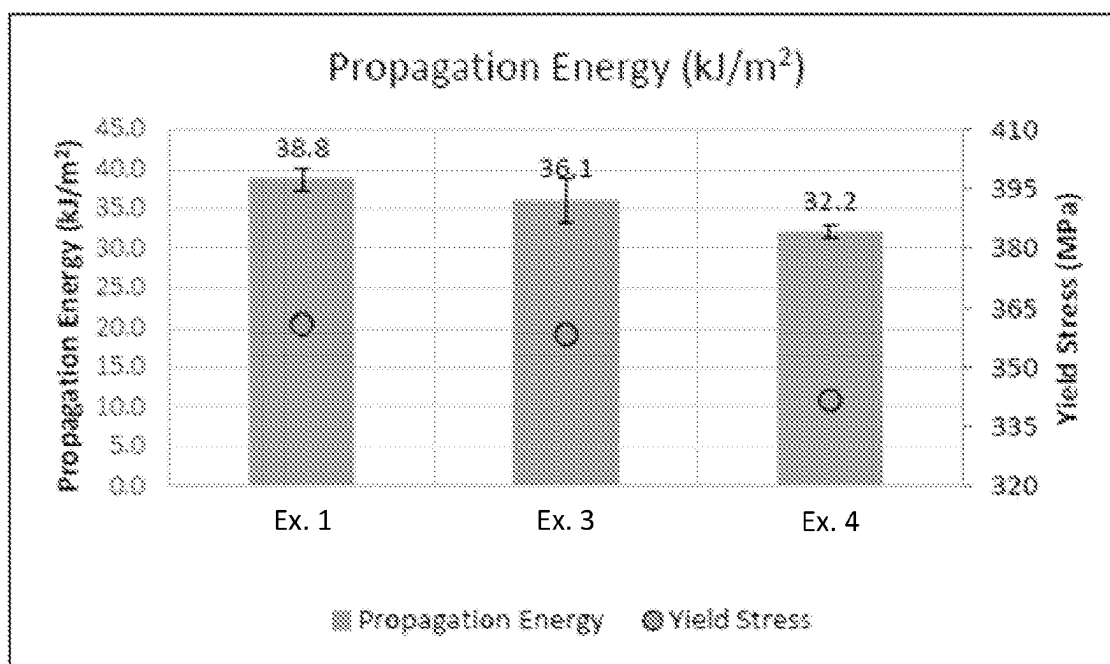
**FIG. 5A****FIG. 5B**

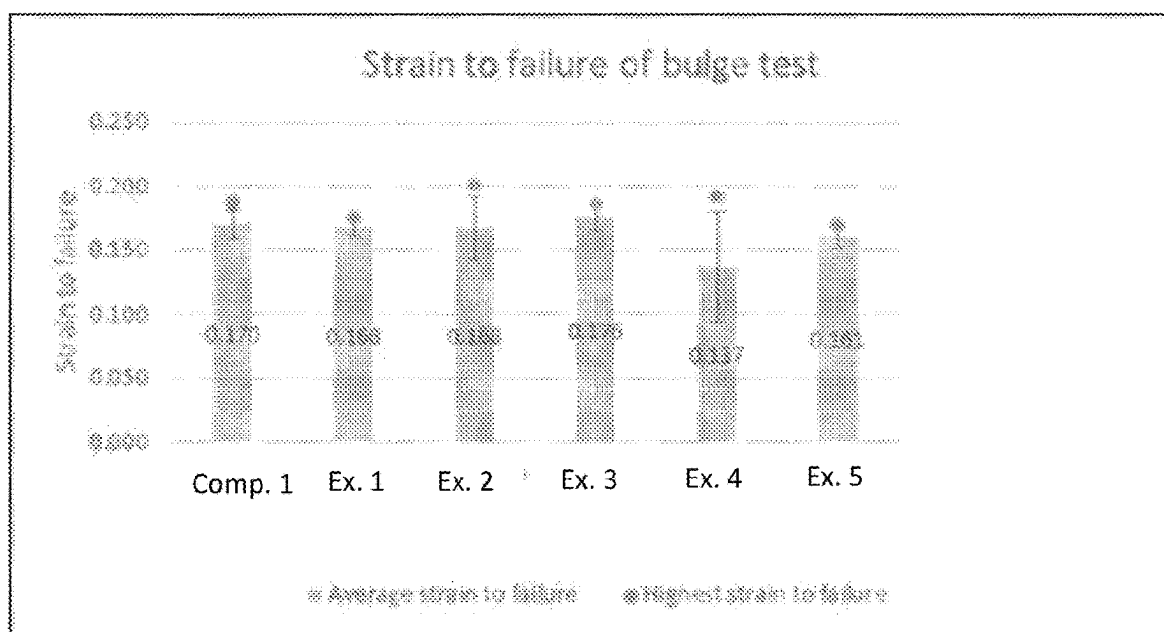
**FIG. 6**

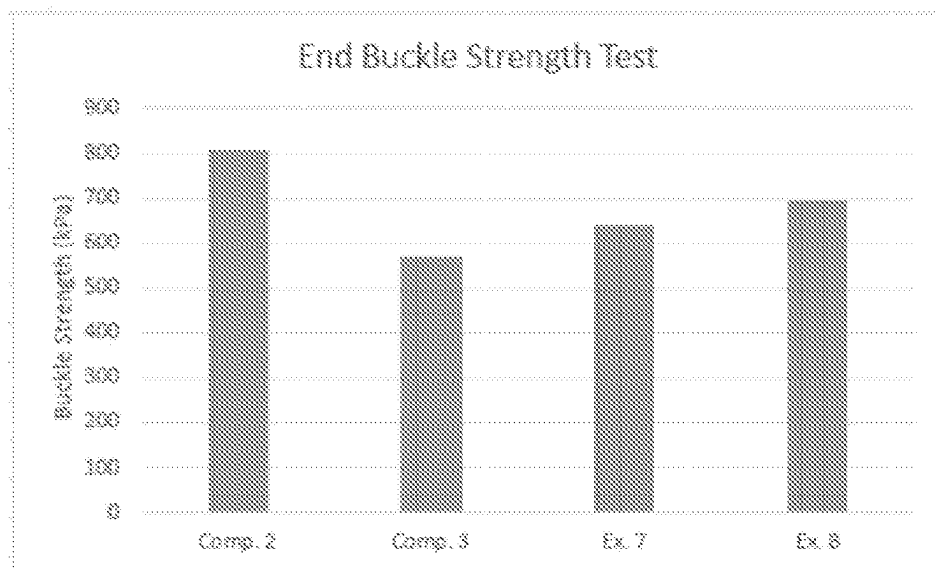
**FIG. 7A****FIG. 7B**

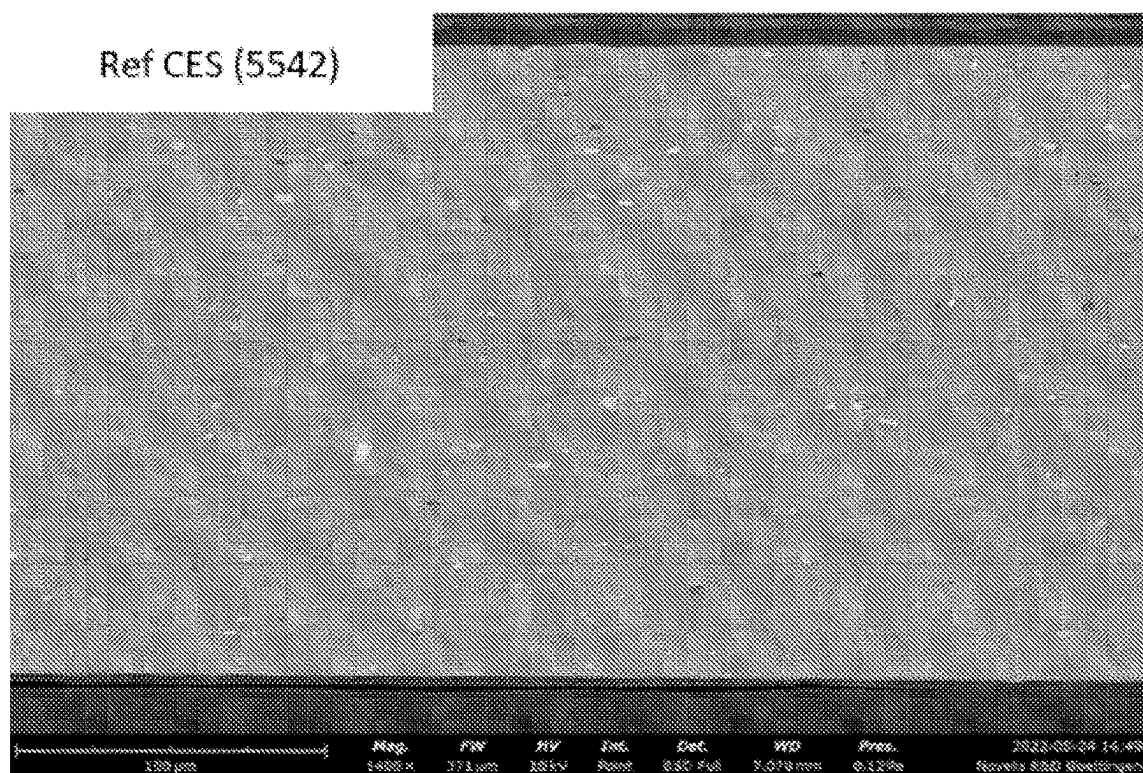
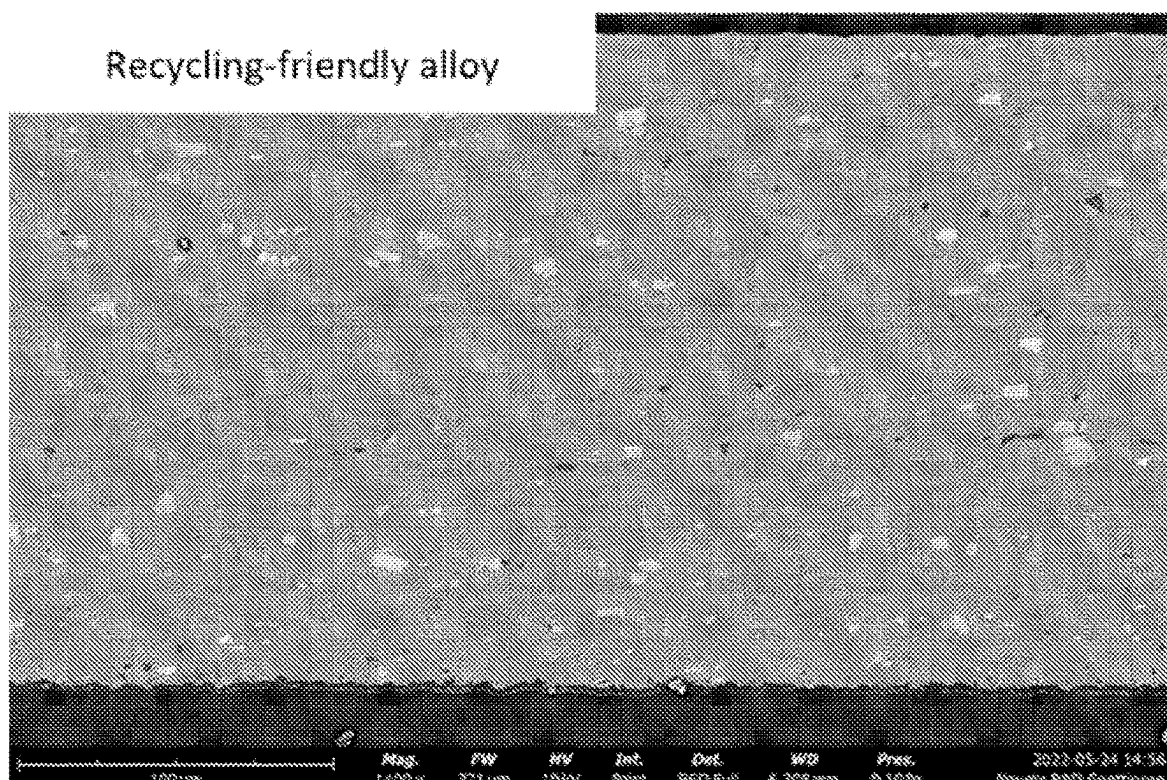
**FIG. 8**

**FIG. 9A****FIG. 9B**

**FIG. 10**

**FIG. 11**

**FIG. 12**

**FIG. 13A****FIG. 13B**

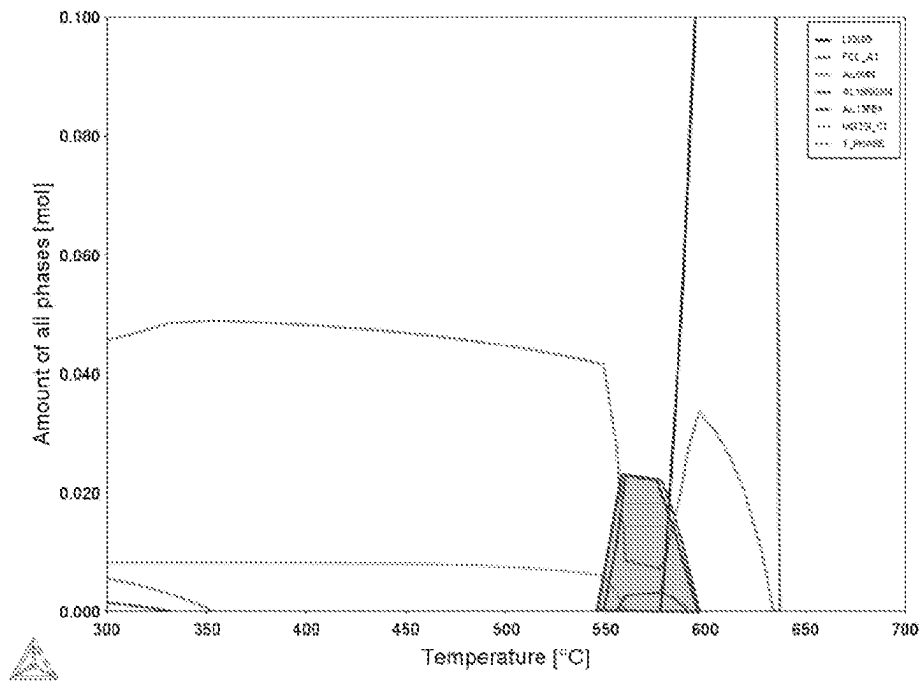


FIG. 14A

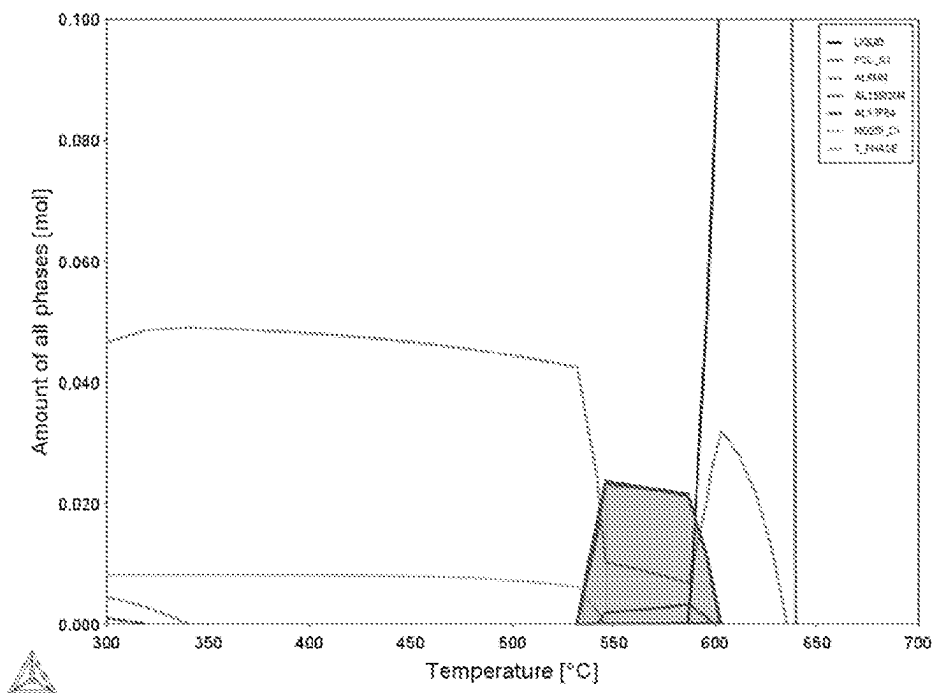


FIG. 14B

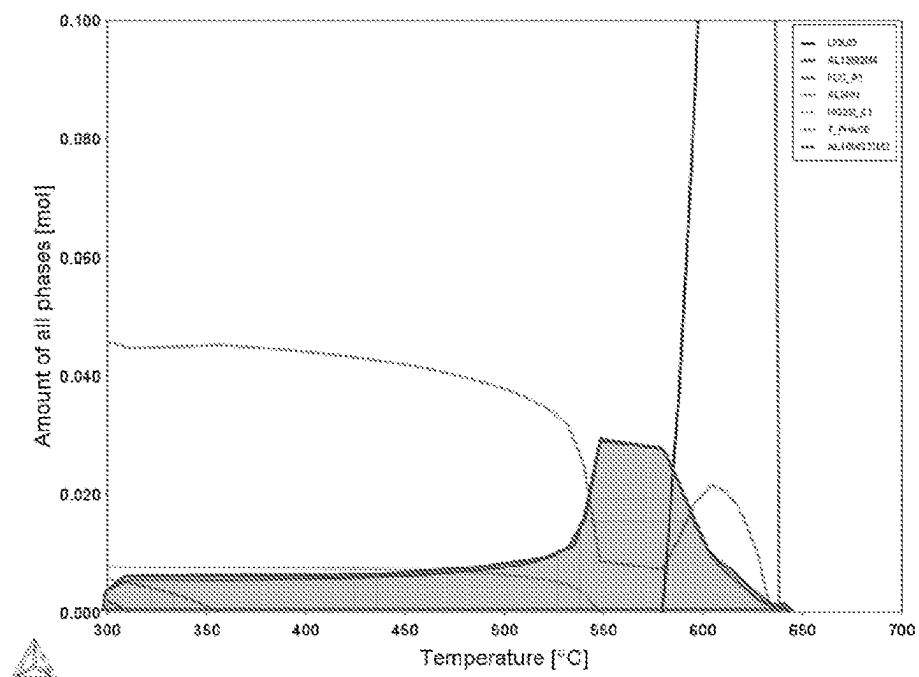


FIG. 14C

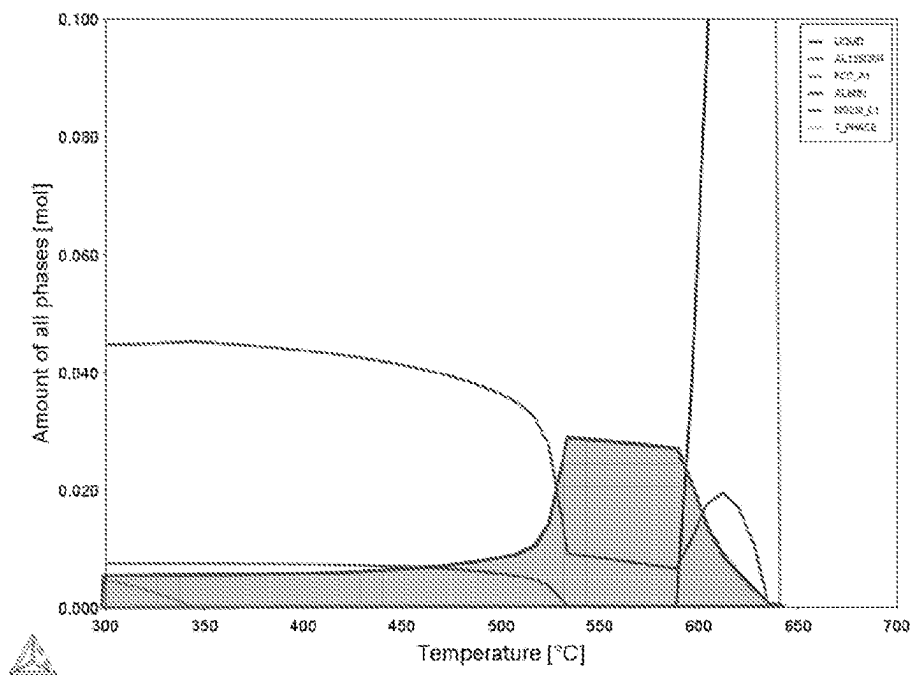


FIG. 14D

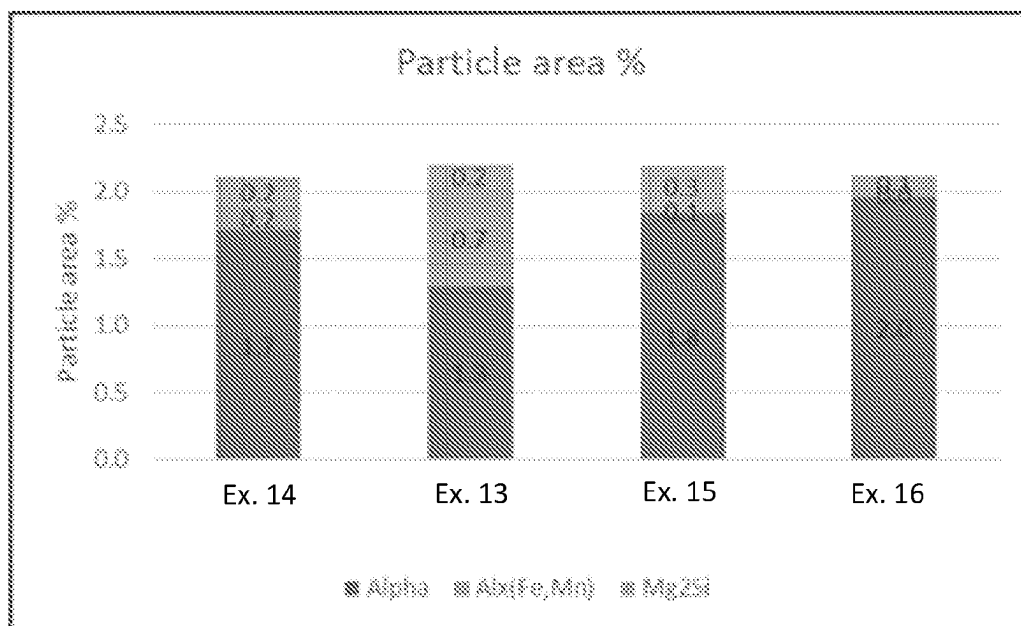


FIG. 15A

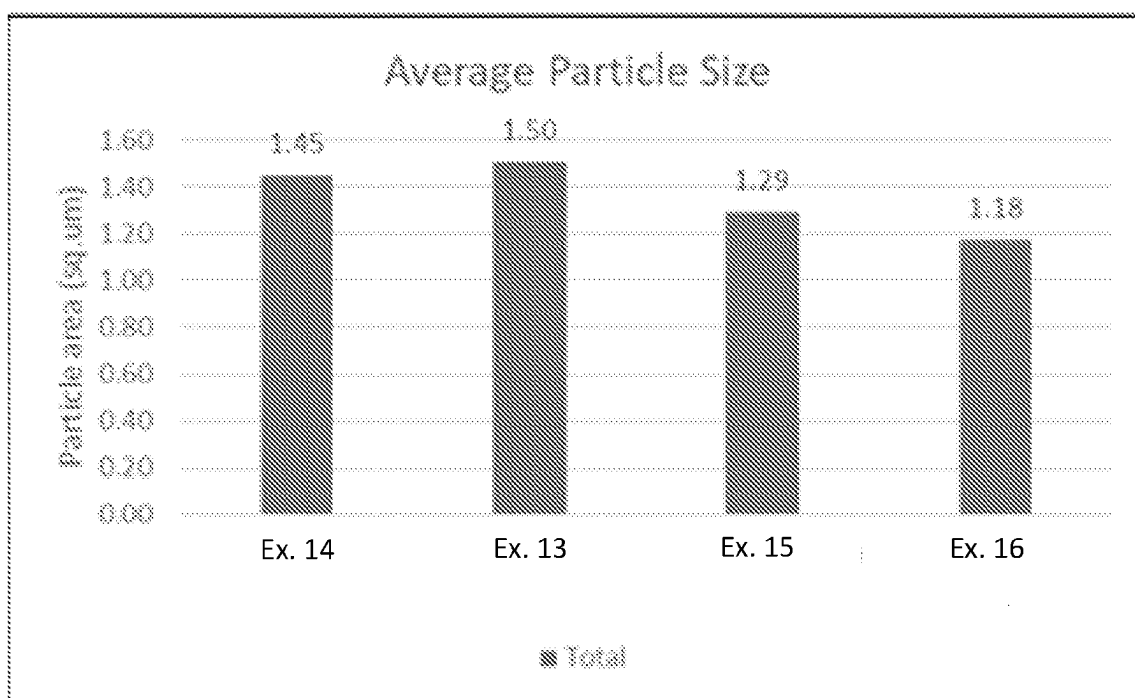


FIG. 15B

HIGH-STRENGTH ALUMINUM ALLOYS FOR CAN END STOCK AND METHODS FOR PREPARING THE SAME

REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of U.S. Provisional Patent Application No. 63/364,163, filed on May 4, 2022, the contents of which is hereby incorporated by reference in its entirety for all intents and purposes.

FIELD

[0002] The present disclosure relates to the fields of metallurgy, aluminum alloys, aluminum fabrication, and related fields. In particular, the present disclosure provides novel aluminum alloys having high amounts of recycled aluminum materials, which can be useful for producing can end stock. The present disclosure also relates to methods of producing novel aluminum alloys from recycled aluminum alloy materials that provides control of the particle size of constituents in the aluminum alloy microstructure for improved mechanical properties of the aluminum alloy.

BACKGROUND

[0003] Can end stock is conventionally made from high-strength aluminum alloys that have good formability properties. The mechanical requirements for aluminum alloys used to produce can end stock are different than the mechanical requirements for can body stock. In general, aluminum alloys for producing can end stock require greater strength than can body stock. As a result, can end stock is often fabricated from an aluminum alloy comprising high amounts of magnesium (Mg). For instance, can end stock may be fabricated from a highly engineered AA5182 aluminum alloy that has a rigidly controlled composition and process for producing the alloy.

[0004] Many aluminum manufacturers use the same alloy for can end stock. Attempts to modify the aluminum alloy composition for can end stock have not been successful primarily because the mechanical properties (e.g., strength and formability) are significantly affected by changes in the aluminum alloy composition. For example, the composition of the AA5182 aluminum alloy is strictly controlled to have a Mg content between 4.0 wt. % and 5.0 wt. %, a manganese (Mn) content between 0.20 wt. % and 0.50 wt. %, a maximum iron (Fe) content of 0.35 wt. %, a maximum silicon (Si) content of 0.20 wt. %, a maximum copper (Cu) content of 0.15 wt. %, and a maximum chromium (Cr) content of 0.10 wt. %. However, recycled material, such as Used Beverage Cans (UBC), are not used to produce can end stock and can body stock because UBC contains two separate aluminum alloys having different aluminum alloy compositions. Specifically, can end stock is typically produced from AA5182 aluminum alloy and can body stock is typically produced from AA3104 aluminum alloy. Due to the two dissimilar aluminum alloys in UBC, there is little commonality in the composition to make new aluminum alloys for can body stock and can end stock. Therefore, if UBC is used to produce new aluminum alloys, there is a need to add primary aluminum and additional alloying elements to adjust the composition to produce can end stock and can body stock. This reduces the circularity of recycled aluminum alloy products for producing new aluminum alloys and requires the addition of primary aluminum, which

decreases the recycled content. Moreover, additional primary aluminum increases the carbon dioxide production and increases costs, leading to environmental harm and high costs.

[0005] Additionally, aluminum alloys produced from a high-content of recycled aluminum materials generally include high amounts of alloying elements. The alloying elements can produce large constituents in the aluminum alloy microstructure that are difficult to breakdown and negatively affect the mechanical properties of the aluminum alloy. For example, high amounts of Si, Fe, and/or Mn, which are common alloying elements in recycled aluminum alloy materials, can produce large constituents in the aluminum alloy microstructure (e.g., $Al_x(Fe,Mn)$) that cannot be sufficiently processed or broken down during aluminum alloy production (e.g., the large constituents cannot be broken down easily during rolling deformation). Therefore, the large constituents remain in the aluminum alloy microstructure of the aluminum alloy product (e.g., sheet) after production. The large constituents in the aluminum alloy microstructure can negatively affect, for example, the formability of the aluminum alloy.

SUMMARY

[0006] Covered embodiments of the present disclosure are defined by the claims, not this summary. This summary is a high-level overview of various aspects of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification, any or all drawings and each claim.

[0007] Described herein are aluminum alloys that provide a more cost-effective and recycled-friendly material alternative to the use of AA5182 alloys for can end stock. In some embodiments, the present disclosure relates to an aluminum alloy including 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 1.3-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some embodiments, the present disclosure relates to an aluminum alloy including 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some aspects, the aluminum alloy includes 0.10-0.30 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-1.0 wt. % Mn, 2.2-5.0 wt. % Mg, up to 0.15 wt. % Cr, up to 0.30 wt. % Zn, up to 0.15 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some aspects, the aluminum alloy includes 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.10 wt. % Cr, up to 0.25 wt. % Zn, up to 0.10 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some aspects, the aluminum alloy includes 0.20-0.35 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.05 wt. % Cr, up to 0.25 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some aspects, the aluminum alloy includes 0.20-0.35 wt. % Si, 0.40-0.60 wt. % Fe, 0.15-0.25 wt. % Cu, 0.60-1.2 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.03 wt. % Cr, up to 0.20 wt. % Zn, up to 0.03 wt. % Ti, up to

0.15 wt. % impurities, and Al. In some aspects, a ratio of Mg:Cu is from 10:1 to 80:1 and a ratio of Mn:Cu is from 2:1 to 15:1. In some aspects, a ratio of Mg:Cu is from 15:1 to 70:1 and a ratio of Mn:Cu is from 3:1 to 12:1. In some aspects, the aluminum alloy has a combined content of Fe and Si of greater than 0.40 wt. %. In some aspects, the aluminum alloy has a combined content of Mg, Mn, and Cu of from 3.5 wt. % to 5.0 wt. %. In some aspects, the aluminum alloy comprises at least 40 wt. % of recycled scrap. In some aspects, the aluminum alloy has less than 30 wt. % of primary aluminum. In some aspects, the aluminum alloy has a propagation energy of at least 30.0 KJ/m² as measured by ASTM B871-1 (2021). In some aspects, the aluminum alloy has a yield strength of at least 340 MPa. In some aspects, aluminum alloy has an ultimate tensile strength of at least 380 MPa. In some aspects, the aluminum alloy has a total elongation of at least 4%. In some aspects, a can end stock comprises any one of the aforementioned aluminum alloys.

[0008] In some embodiments, a method for producing an aluminum alloy is provided. The method includes casting an aluminum alloy to form a cast product, wherein the aluminum alloy comprises 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al; homogenizing the cast product; hot rolling the cast product to produce a hot rolled product; cold rolling the hot rolled product to produce a final gauge rolled product; and optionally annealing the final gauge rolled product. In some aspects, the method includes lacquering and curing the final gauge rolled product. In some aspects, the homogenization step includes a first homogenization step and a second homogenization step. In some aspects, the first homogenization step includes soaking the cast product at a temperature from 375° C. to 450° C. for 0.5 hours to 5 hours. In some aspects, the second homogenization step includes soaking the cast product at a temperature from 450° C. to 550° C. for 0.01 hours to 5 hours. In some aspects, the aluminum alloy includes 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.10 wt. % Cr, up to 0.25 wt. % Zn, up to 0.10 wt. % Ti, up to 0.15 wt. % of impurities, and Al. In some aspects, a ratio of Mg:Cu is from 15:1 to 70:1 and a ratio of Mn:Cu is from 3:1 to 12:1. In some aspects, a metal product is prepared by the aforementioned method. In some aspects, the metal product is can end stock.

[0009] In some embodiments, the present disclosure provides an aluminum alloy including 0.01-0.60 wt. % Si, 0.01-0.80 wt. % Fe, 0.05-0.30 wt. % Cu, 0.80-1.40 wt. % Mn, 1.3-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al. In some embodiments, the aluminum alloy includes 0.10-0.50 wt. % Si, 0.20-0.70 wt. % Fe, 0.11-0.30 wt. % Cu, 0.80-1.00 wt. % Mn, 1.5-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al. In some embodiments, the aluminum alloy includes 0.22-0.32 wt. % Si, 0.50-0.65 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al. In some embodiments, the aluminum alloy includes 0.25-0.32 wt. % Si, 0.45-0.55 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, 0.01-0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. %

Ti, up to 0.15 wt. % of impurities, and remainder Al. In some embodiments, the aluminum alloy includes 0.25-0.35 wt. % Si, 0.45-0.55 wt. % Fe, 0.16-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 3.0-4.0 wt. % Mg, 0.05-0.15 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al. In some embodiments, the aluminum alloy comprises 0.05-0.20 wt. % Cr. In some embodiments, the aluminum alloy has a ratio of Mg:Cr is from 20:1 to 70:1. In some embodiments, the aluminum alloy has a combined content of Si, Cr, and Cu, is greater than 0.35 wt. %. In some embodiments, the aluminum alloy has a combined content of Fe and Si is greater than 0.40 wt. %. In some embodiments, particles in the aluminum alloy microstructure have a particle size, measured by area, of 1.45 μm² or less. In some embodiments, the aluminum alloy comprises at least 40 wt. % of recycled scrap. In some embodiments, the aluminum alloy comprises less than 30 wt. % of primary aluminum. In some embodiments, the aluminum alloy has a yield strength of at least 340 MPa. In some embodiments, the aluminum alloy has an ultimate tensile strength of at least 380 MPa. In some embodiments, the aluminum alloy has the aluminum alloy has a total elongation of at least 4%. In some embodiments a can end stock comprises any of the aforementioned aluminum alloys.

[0010] In some embodiments, a method for producing an aluminum alloy is provided. The method includes casting an aluminum alloy to form a cast product, wherein the aluminum alloy comprises 0.01-0.60 wt. % Si, 0.01-0.80 wt. % Fe, 0.05-0.30 wt. % Cu, 0.80-1.40 wt. % Mn, 1.3-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al, homogenizing the cast product, wherein homogenizing the cast product produces alpha phase particles; hot rolling the cast product to produce a hot rolled product; cold rolling the hot rolled product to produce a final gauge rolled product; and optionally annealing the final gauge rolled product. In some embodiments, the aluminum alloy comprises 0.05-0.20 wt. % Cr, and wherein a ratio of Mg:Cr is from 20:1 to 70:1. In some embodiments, the homogenization step comprises heating and soaking the cast product at a temperature from 450° C. to 570° C., wherein homogenization is configured to transform large particles to alpha phase particles. In some embodiments, the cast product is soaked at the homogenization temperature for up to 10 hours. In some embodiments, the aluminum alloy includes a particle area % of alpha phase particles, as measured by volume, of 1.5% or greater after homogenization. In some embodiments, the aluminum alloy comprises 0.22-0.32 wt. % Si, 0.50-0.65 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al. **[0011]** Further aspects, objects, and advantages will become apparent upon consideration of the detailed description and figures that follow.

BRIEF DESCRIPTION OF THE FIGURES

[0012] FIG. 1 provides a graph of the yield stress, ultimate tensile strength, and total elongation of example aluminum alloys according to some embodiments described herein.

[0013] FIG. 2 provides a graph of the yield stress (MPa) when measured in a longitudinal (L) direction, a transverse (T) direction, and in a diagonal (D) direction, each respective to the rolling direction, of example aluminum alloys according to some embodiments described herein.

[0014] FIGS. 3A and 3B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of the aluminum alloy as it relates to the concentration of Fe and Si in the example aluminum alloys.

[0015] FIG. 4 provides a graph of the propagation energy and yield stress of example alloys provided in FIGS. 3A and 3B as measured according to ASTM B873-1 (2021).

[0016] FIGS. 5A and 5B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of the aluminum alloy as it relates to the concentration of Mn in the example aluminum alloys.

[0017] FIG. 6 provides a graph of the propagation energy and yield stress of example alloys provided in FIGS. 5A and 5B as measured according to ASTM B873-1 (2021).

[0018] FIGS. 7A and 7B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of the aluminum alloy as it relates to the concentration of Mg in the example aluminum alloys.

[0019] FIG. 8 provides a graph of the propagation energy and yield stress of example alloys provided in FIGS. 7A and 7B as measured according to ASTM B873-1 (2021).

[0020] FIGS. 9A and 9B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of the aluminum alloy as it relates to the concentration of Mg and Mn in the example aluminum alloys.

[0021] FIG. 10 provides a graph of the propagation energy and yield stress of example alloys provided in FIGS. 9A and 9B as measured according to ASTM B873-1 (2021).

[0022] FIG. 11 provides a graph of the strain to failure for bulge tests of the example alloys as measured according to ISO 16808 (2021).

[0023] FIG. 12 provides a graph of the buckle strength of example alloys.

[0024] FIGS. 13A and 13B provide scanning electron microscopy (SEM) images of a microstructure of a AA5182 aluminum alloy (FIG. 13A) and an example aluminum alloy produced from high amounts of recycled aluminum materials (FIG. 13B).

[0025] FIGS. 14A-14D provide graphs representing the particles in the aluminum alloy microstructure as a function of the homogenization temperature for Example Alloy 9 (FIG. 14A), Example Alloy 10 (FIG. 14B), Example Alloy 11 (FIG. 14C), and Example Alloy 12 (FIG. 14D), based on thermodynamic equilibrium calculations performed on Thermo-Calc.

[0026] FIGS. 15A and 15B provide graphs of the particle area % (FIG. 15A) and the average particle size (FIG. 15B), as measured by area, of particles in the aluminum alloy microstructure for Example Alloys 13-16.

DETAILED DESCRIPTION

[0027] Described herein are novel aluminum alloys which exhibit high strength and formability. Surprisingly, the aluminum alloys described herein exhibit high strength and formability despite having a lower Mg content than traditional AA5182 aluminum alloys used to produce can end stock. The aluminum alloys described herein incorporate higher amounts of recycled aluminum materials and less primary aluminum, as compared to traditional aluminum alloys used to produce can end stock, and still maintain good

mechanical properties for can end stock. For example, the aluminum alloys described herein may include more than 40% recycled aluminum alloy and less than 30% primary aluminum than traditional aluminum alloys used to produce can end stock, and still exhibit equivalent properties to AA5182 aluminum alloys. The aluminum alloy composition described throughout provides a cost-effective alternative to the use of AA5182 aluminum alloys for can end stock.

[0028] Conventional AA5182 aluminum alloys for producing can end stock require a strictly controlled composition to meet the minimum strength requirements for can end stock while still maintaining formability to produce complex geometries. In general, greater strength is required for aluminum alloys used to produce can end stock compared to can body stock, which has dictated that such can end stock be fabricated from an aluminum alloy including high amounts of Mg, such as AA5182 aluminum alloy. This limits the amount of recycled aluminum material that can be used to produce AA5182 aluminum alloy. For example, AA5182 aluminum alloy cannot be produced from high amounts of used beverage cans because AA5182 aluminum alloy includes lower amounts of Fe, Si, Mn, and Cu compared to used beverage cans. Used beverage cans include a mixture of two different aluminum alloys for can body stock and can end stock. The aluminum alloy used for can end stock is typically AA5182 aluminum alloy and the aluminum alloy used for can body stock is typically AA3104 aluminum alloy. AA3104 aluminum alloy includes lower amounts of Mg and higher amounts of Fe, Si, Mn, and Cu compared to AA5182 aluminum alloy. Due to the discrepancy between the aluminum alloy composition of AA3104 aluminum alloy and AA5182 aluminum alloy, used beverage cans have an aluminum alloy composition between the compositions for AA3104 aluminum alloy and AA5182 aluminum alloy. Therefore, in order to produce can end stock with large amounts of recycled materials, such as used beverage cans, it requires the use of more alloying elements (e.g., Mg addition) and primary aluminum to produce AA5182 aluminum alloy, which significantly increases the cost of materials. This limits the amount of the recycled aluminum materials that can be used to produce can end stock.

[0029] The novel aluminum alloy described herein can utilize higher amounts of recycled aluminum materials and achieve properties similar to AA5182 aluminum alloys. Specifically, the aluminum alloy described herein can tolerate higher amounts of Fe, Si, Mn, and/or Cu compared to AA5182 aluminum alloy and still achieve good strength and formability properties. Additionally, the aluminum alloys described herein may include lower amounts of Mg that can be compensated by the addition of elements such as Mn and Cu to balance the strength. The composition of the aluminum alloy described herein reduces the compositional gap between can body stock and can end stock to lower the amount of primary aluminum and reduce the addition of alloying elements (e.g., Mg). By reducing the compositional gap between aluminum alloys for can body stock and can end stock, more recycled aluminum alloy, such as used beverage cans, may be used to produce aluminum alloys for can end stock. For example, the aluminum alloy described herein can be produced from at least 40 wt. % of recycled scrap and less than 30 wt. % of primary aluminum.

[0030] Additionally, the aluminum alloy composition produces can end stock having similar properties to conventional AA5182 aluminum alloys, allowing can manufactur-

ers to produce the aluminum alloy with no changes to their existing methods. In some embodiments, higher amounts of used beverage cans may be used with the aluminum alloy described herein, thereby reducing the amount of primary aluminum needed and reducing the total cost and maintaining equivalent or better rolling productivity.

[0031] The aforementioned aluminum alloy composition has processing advantages despite having high amounts of recycled aluminum alloy, which is a common problem for using high amounts of recycled aluminum alloy for new aluminum alloys. Specifically, the amount of Mg in the aluminum alloy composition can be naturally reduced to about 1 wt. % Mg during a re-melting process via fluxing, degassing, and dross treatment. Therefore, the final molten aluminum alloy composition can be very similar to AA3104 aluminum alloy. The aluminum alloys described herein can be reused to make can body stock. Additionally, with the proper adjustment of the Mg content, the aluminum alloys described herein can also be recycled to produce can end stock. Beneficially, this allows for closed-loop recycling of Used Beverage Cans (UBC) produced from the aluminum alloy composition described herein. In this way, UBC scrap can be continually reused in a closed loop system to produce the aluminum alloys without major changes to the alloying elements. That is, the aluminum alloy products (e.g., UBC) made from the aluminum alloy composition described herein can be used to produce new aluminum alloys for can end stock or can body stock. Moreover, because Fe, Si, Cu, and Mn level in UBC scrap is similar to the aluminum alloy composition described herein, only additional Mg needs to be added to the aluminum alloy to produce can end stock from UBC.

[0032] In some embodiments, the present disclosure relates to an aluminum alloy that has a similar composition to AA3104 aluminum alloy (except for Mg) and exhibits properties similar to AA5182 aluminum alloy (e.g., buckle strength). For example, the aluminum alloy may include a Fe, Si, Cu, and Mn content similar to AA3104 aluminum alloy and may include a Mg content greater than 2.0 wt. %. The aluminum alloys can include high amounts of recycled material and achieve properties similar to AA5182 aluminum alloy. In some embodiments, the aluminum alloys described herein may have a composition similar to AA3104 aluminum alloy and may include a Mg content from 2.0 wt. % to 4.0 wt. % and exhibit a buckle strength greater than 90 psi. Beneficially, the aluminum alloy includes similar amounts of Fe, Si, Cu, and Mn as AA3104 aluminum alloy, which is used for UBC. Therefore, UBC scrap can be used almost entirely to produce the aluminum alloy, which eliminates the need of diluting the aluminum alloy with primary aluminum or adding alloying elements for hardening. This provides an aluminum alloy that can include higher amounts of UBC scrap and less primary aluminum and minimal or no additional alloying elements other than Mg. Advantageously, the aluminum alloy may include a Mg content to meet the strength requirements for can end stock. If UBC scrap is used to produce aluminum alloys without Mg addition, the strength of the resulting aluminum alloy may be substantially lower and may result in insufficient buckle strength (e.g., shell buckle strength and can end buckle strength). The aluminum alloys described herein may include higher amounts of the Mg compared to AA3104 aluminum alloy to meet the minimum strength requirements for can end stock. In some embodiments, the aluminum

alloy described herein may have a composition similar to AA3104 aluminum alloy and may include a Mg content greater than 1.3 wt. % (e.g., 1.3 wt. % to 2.0 wt. % Mg). This aluminum alloy may be used in applications that have lower strength requirements.

[0033] To maintain a high level of recycled content in the aluminum alloy compositions described herein, the aluminum alloy composition can be similar to AA3104 aluminum alloy. For example, Mg can be the primary alloy element that is added to the aluminum alloy composition and the remaining alloy elements may be similar to AA3104 aluminum alloy. Therefore, other hardening alloying elements (e.g., Mn and Cu) may not need to be added to the aluminum alloy composition. This is beneficial because hardening alloying elements (e.g., Mn and Cu) do not oxidize during the re-melting process, thereby affecting the recyclability of the can body aluminum alloy or UBC composition. By maintaining aluminum alloy compositions similar to AA3104 aluminum alloy, the re-melting process may be simplified and reduces the need to change the aluminum alloy during fabrication. The aluminum alloy compositions described herein contain higher levels of recycled content and less primary aluminum while also demonstrating mechanical properties similar to current aluminum alloy compositions for can end stock (e.g., similar to AA5182 aluminum alloy).

[0034] In some embodiments, the present disclosure provides novel methods of making aluminum alloys produced from high amounts of recycled aluminum alloy materials (e.g., compared to AA5182 aluminum alloy). Beneficially, the aluminum alloy composition described herein includes a specific combination of alloy elements to produce alpha phase particles during a carefully controlled homogenization practice. For example, by reducing the Mg content and/or adding Cr (e.g., up to 0.20 wt. %) to the aluminum alloy composition, the aluminum alloy can be homogenized at a wider range of homogenization temperatures to produce a higher volume fraction of alpha phase particles (e.g., compared to other aluminum alloys that does not include the same combination of alloying elements). The homogenization temperature and control of the aluminum alloy composition maximizes formation of alpha particles. For example, the aluminum alloys described herein include a higher volume fraction of alpha particles compared to aluminum alloys that are also produced from high amounts of recycled aluminum alloy materials. The alpha phase particles include phases (e.g., eutectic phases) that can be broken down during hot rolling into smaller particles thereby resulting in an aluminum alloy having good formability and mechanical properties.

[0035] Aluminum alloys that are produced from a high content of recycled aluminum alloy materials include large constituents in the aluminum alloy microstructure that are difficult to breakdown and negatively affect the mechanical properties of the aluminum alloy. For example, high amounts of Si, Fe, and/or Mn, which are common alloying elements in recycled aluminum alloy materials, can produce large constituents in the aluminum alloy microstructure (e.g., Al(Fe,Mn)) that cannot be sufficiently processed or broken-down during aluminum alloy production. The constituent particle size and the number density of constituents in the aluminum alloy microstructure determine, in part, the formability and performance of the aluminum alloy. Therefore, particle size control is beneficial for producing aluminum alloys having desired mechanical properties.

[0036] The aluminum alloys described herein having a carefully controlled amount of Si, Mg, Cu, and Cr and produced according to the methods described herein provides an aluminum alloy microstructure including higher amounts of alpha phase particles and less amounts of large constituents compared to other aluminum alloys produced from recycled aluminum alloy materials. Specifically, the aluminum alloy has a microstructure including a high number density of small particles by modifying the alloying elements and homogenizing at temperatures where alpha phase transformation occurs. The aforementioned aluminum alloys have a wide range of homogenization temperatures that transforms large particles such as to $Al_x(Fe,Mn)$ alpha phase particles. Additionally, the aluminum alloy is homogenized at homogenization temperatures that transforms large constituents into alpha phase particles. In some embodiments, the aforementioned aluminum alloys can be homogenized at a temperature range from 45° C. to 570° C. to provide a higher volume fraction of alpha phase particles compared to other aluminum alloys produced from recycled aluminum alloy materials. The alpha phase particles produced during homogenization can be broken down into smaller particles during hot rolling, thereby significantly improving formability.

Definitions and Descriptions

[0037] As used herein, the terms “invention,” “the invention,” “this invention” and “the present invention” are intended to refer broadly to all of the subject matter of this patent application and the claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below.

[0038] In this description, reference is made to alloys identified by aluminum industry designations, such as “series” or “5xxx.” For an understanding of the number designation system most commonly used in naming and identifying aluminum and its alloys, see “International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys,” or “Registration Record of Aluminum Association Alloy Designations and Chemical Compositions Limits for Aluminum Alloys in the Form of Castings and Ingot,” both published by The Aluminum Association.

[0039] As used herein, the meaning of “a,” “an,” or “the” includes singular and plural references unless the context clearly dictates otherwise.

[0040] As used herein, a plate generally has a thickness of greater than about 15 mm. For example, a plate may refer to an aluminum product having a thickness of greater than about 15 mm, greater than about 20 mm, greater than about 25 mm, greater than about 30 mm, greater than about 35 mm, greater than about 40 mm, greater than about 45 mm, greater than about 50 mm, or greater than about 100 mm.

[0041] As used herein, a shate (also referred to as a sheet plate) generally has a thickness of from about 4 mm to about 15 mm. For example, a shate may have a thickness of about 4 mm, about 5 mm, about 6 mm, about 7 mm, about 8 mm, about 9 mm, about 10 mm, about 11 mm, about 12 mm, about 13 mm, about 14 mm, or about 15 mm.

[0042] As used herein, a sheet generally refers to an aluminum product having a thickness of less than about 4 mm (e.g., less than 3 mm, less than 2 mm, less than 1 mm, less than 0.5 mm, less than 0.3 mm, or less than 0.1 mm). For

example, a sheet may have a thickness of about 0.1 mm, about 0.2 mm, about 0.3 mm, about 0.4 mm, about 0.5, about 0.6 mm about 0.7 mm, about 0.8 mm, about 0.9 mm, about 1 mm, about 1.1 mm, about 1.2 mm, about 1.3 mm, about 1.4 mm, about 1.5 mm, about 1.6 mm, about 1.7 mm, about 1.8 mm, about 1.9 mm, about 2 mm, about 2.1 mm, about 2.2 mm, about 2.3 mm, about 2.4 mm, about 2.5 mm, about 2.6 mm about 2.7 mm, about 2.8 mm, about 2.9 mm, about 3 mm, about 3.1 mm, about 3.2 mm, about 3.3 mm, about 3.4 mm, about 3.5 mm, about 3.6 mm, about 3.7 mm, about 3.8 mm, about 3.9 mm, or about 4 mm.

[0043] As used herein, formability refers to the ability of a material to undergo deformation into a desired shape without fracturing, tearing-off, necking, earing, or shaping errors such as wrinkling, spring-back, or galling occurring. In engineering, formability may be classified according to deformation modes. Examples of deformation modes include drawing, stretching, bending, and stretch-flanging.

[0044] As used herein, primary aluminum refers to an aluminum material including about at least 99.7 wt. % aluminum. Primary aluminum is produced from the prime transformation of raw material into aluminum (e.g., processing of bauxite into alumina and electrolysis of alumina into aluminum).

[0045] As used herein, yield stress (also referred to as yield strength) refers to the point at which an aluminum alloy begins to plastically deform and can no longer return to its original state.

[0046] Reference may be made in this application to alloy temper or condition. For an understanding of the alloy temper descriptions most commonly used, see “American National Standards (ANSI) H35 on Alloy and Temper Designation Systems.” An F condition or temper refers to an aluminum alloy as fabricated. An O condition or temper refers to an aluminum alloy after annealing. An Hxx condition or temper, also referred to herein as an H temper, refers to a non-heat treatable aluminum alloy after cold rolling with or without thermal treatment (e.g., annealing). Suitable H tempers include HX1, HX2, HX3 HX4, HX5, HX6, HX7, HX8, or HX9 tempers. A T1 condition or temper refers to an aluminum alloy cooled from hot working and naturally aged (e.g., at room temperature). A T2 condition or temper refers to an aluminum alloy cooled from hot working, cold worked and naturally aged. A T3 condition or temper refers to an aluminum alloy solution heat treated, cold worked, and naturally aged. A T4 condition or temper refers to an aluminum alloy solution heat treated and naturally aged. A T5 condition or temper refers to an aluminum alloy cooled from hot working and artificially aged (at elevated temperatures). A T6 condition or temper refers to an aluminum alloy solution heat treated and artificially aged. A T7 condition or temper refers to an aluminum alloy solution heat treated and artificially overaged. A T8x condition or temper refers to an aluminum alloy solution heat treated, cold worked, and artificially aged. A T9 condition or temper refers to an aluminum alloy solution heat treated, artificially aged, and cold worked. A W condition or temper refers to an aluminum alloy after solution heat treatment.

[0047] As used herein, the meaning of “room temperature” can include a temperature of from about 15° C. to about 30° C., for example about 15° C., about 16° C., about 17° C., about 18° C., about 19° C., about 20° C., about 21°

C., about 22° C., about 23° C., about 24° C., about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., or about 30° C.

[0048] All ranges disclosed herein are to be understood to encompass both endpoints and any and all subranges subsumed therein. For example, a stated range of “1 to 10” should be considered to include any and all subranges between (and inclusive of) the minimum value of 1 and the maximum value of 10; that is, all subranges beginning with a minimum value of 1 or more, e.g. 1 to 6.1, and ending with a maximum value of 10 or less, e.g., 5.5 to 10.

[0049] The following aluminum alloys are described in terms of their elemental composition in weight percentage (wt. %) based on the total weight of the alloy. In certain examples of each alloy, the remainder is aluminum, with a maximum wt. % of 0.15% for the sum of the impurities.

Alloy Compositions

[0050] Aluminum alloy properties are partially determined by the composition of the aluminum alloy. In certain aspects, the alloy composition may influence or even determine whether the alloy will have properties adequate for a desired application.

[0051] The alloys described herein are novel aluminum alloys. The alloys exhibit high strength and high formability (e.g., excellent elongation and forming properties), while including higher amounts of recycled aluminum alloys. The properties of the alloys are achieved at least in part due to the elemental composition of the alloys. In some cases, the novel aluminum alloys described herein can include a lower Mg content and higher levels of Si and Fe compared to conventional AA5182 aluminum alloys, as further described below.

[0052] In some examples, an aluminum alloy as described herein can have the following elemental composition as provided in Table 1.

TABLE 1

Element	Weight Percentage (wt. %)
Si	0.10-0.35
Fe	0.20-0.60
Cu	0.05-0.25
Mn	0.25-1.20
Mg	2.0-5.0
Cr	Up to 0.20
Zn	Up to 0.30
Ti	Up to 0.20
Others	0-0.05 (each) 0-0.15 (total)
Al	

[0053] In some examples, the aluminum alloy as described herein can have the following elemental composition as provided in Table 2.

TABLE 2

Element	Weight Percentage (wt. %)
Si	0.10-0.30
Fe	0.20-0.50
Cu	0.05-0.25
Mn	0.30-1.0
Mg	2.2-5.0
Cr	Up to 0.15

TABLE 2-continued

Element	Weight Percentage (wt. %)
Zn	Up to 0.30
Ti	Up to 0.15
Others	0-0.05 (each) 0-0.15 (total)
Al	

[0054] In some examples, the aluminum alloy as described herein can have the following elemental composition as provided in Table 3.

TABLE 3

Element	Weight Percentage (wt. %)
Si	0.10-0.25
Fe	0.20-0.50
Cu	0.05-0.25
Mn	0.30-0.90
Mg	2.5-5.0
Cr	Up to 0.10
Zn	Up to 0.25
Ti	Up to 0.10
Others	0-0.05 (each) 0-0.15 (total)
Al	

[0055] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 4.

TABLE 4

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.20-0.50
Cu	0.05-0.25
Mn	0.30-0.90
Mg	2.5-5.0
Cr	Up to 0.05
Zn	Up to 0.25
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)
Al	

[0056] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 5.

TABLE 5

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.40-0.60
Cu	0.15-0.25
Mn	0.60-1.20
Mg	2.0-4.0
Cr	Up to 0.03
Zn	Up to 0.20
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
Al	

[0057] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 6.

TABLE 6

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.20-0.50
Cu	0.05-0.25
Mn	0.30-0.90
Mg	2.5-5.0
Cr	Up to 0.20
Zn	Up to 0.25
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)

A1

[0058] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 7.

TABLE 7

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.20-0.50
Cu	0.05-0.25
Mn	0.30-0.90
Mg	2.0-3.0
Cr	0.05-0.20
Zn	Up to 0.25
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)

A1

[0059] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 8.

TABLE 8

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.40-0.60
Cu	0.15-0.25
Mn	0.60-1.20
Mg	2.0-4.0
Cr	Up to 0.20
Zn	Up to 0.20
Ti	Up to 0.03
Others	0- 0.05 (each) 0-0.15 (total)

A1

[0060] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 9.

TABLE 9

Element	Weight Percentage (wt. %)
Si	0.20-0.35
Fe	0.40-0.60
Cu	0.15-0.25
Mn	0.60-1.20
Mg	2.0-4.0
Cr	0.05 to 0.20
Zn	Up to 0.20
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)

A1

[0061] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 10.

TABLE 10

Element	Weight Percentage (wt. %)
Si	0.25-0.35
Fe	0.50-0.60
Cu	0.15-0.25
Mn	0.80-1.00
Mg	2.50-3.00
Cr	Up to 0.20
Zn	Up to 0.20
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)

A1

Silicon (Si)

[0062] In some examples, the aluminum alloy described herein includes Si in an amount of from 0.10% to 0.35% (e.g., from 0.10% to 0.30%, from 0.10% to 0.25%, or from 0.20% to 0.35%) based on the total weight of the alloy. For example, the alloy can include 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, 0.30%, 0.31%, 0.32%, 0.33%, 0.34% or 0.35% Si. All expressed in wt. %. In some embodiments, an aluminum alloy composition including less than 0.10 wt. % Si may limit the amount of recycled aluminum material that can be used in the aluminum alloy composition. For example, AA3104 aluminum alloy used for can body stock typically includes 0.6 wt. % Si. In some embodiments, an aluminum alloy composition including greater than 0.35 wt. % Si may form coarse Mg_2Si particles in the aluminum alloy microstructure. The coarse Mg_2Si particles in the aluminum alloy microstructure may reduce formability and strength of the aluminum alloy.

Iron (Fe)

[0063] In some examples, the aluminum alloy described herein also includes Fe in an amount of from 0.20% to 0.60% (e.g., from 0.20% to 0.50%, from 0.20% to 0.35%, or from 0.40% to 0.60%) based on the total weight of the alloy. For example, the alloy can include 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, 0.30%, 0.31%, 0.32%, 0.33%, 0.34%, 0.35%, 0.36%, 0.37%, 0.38%, 0.39%, 0.40%, 0.41%, 0.42%, 0.43%, 0.44%, 0.45%, 0.46%, 0.47%, 0.48%, 0.49%, 0.50%, 0.51%, 0.52%, 0.53%, 0.54%, 0.55%, 0.56%, 0.57%, 0.58%, 0.59%, or 0.60% Fe. All expressed in wt. %. In some embodiments, an aluminum alloy composition including less than 0.20 wt. % Fe may result in processing defects. For example, the aluminum alloy may have poor runnability due to excess die build up. Runnability refers to whether an aluminum alloy includes defects or jams during the process of producing the aluminum alloy. Additionally, including less than 0.20 wt. % Fe in the aluminum alloy composition may limit the amount of recycled aluminum materials that can be used in the aluminum alloy. In some embodiments, an aluminum alloy composition including greater than 0.50 wt. % Fe may result in poor formability due to high amounts of Fe-containing intermetallic particles in the aluminum alloy microstructure.

Copper (Cu)

[0064] In some examples, the aluminum alloy described herein includes Cu in an amount of from 0.05 to 0.25% (e.g., from 0.15% to 0.25%) based on the total weight of the alloy. For example, the alloy can include 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, or 0.25% Cu. All expressed in wt. %. As discussed herein, the aluminum alloy may include 0.05 wt. % to 0.25 wt. % Cu to compensate for the reduced content of Mg to strengthen the aluminum alloy. In some embodiments, aluminum alloys including less than 0.05 wt. % Cu may lead to insufficient strength properties. In some embodiments, an aluminum alloy including greater than 0.25 wt. % Cu may lead to excess strength, poor formability, and susceptibility to corrosion.

Manganese (Mn)

[0065] In some examples, the aluminum alloy described herein can include Mn in an amount from 0.25% to 1.20% (e.g., from 0.30% to 1.0%, from 0.30% to 0.90%, or from 0.60% to 1.20%) based on the total weight of the alloy. For example, the alloy can include 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, 0.30%, 0.31%, 0.32%, 0.33%, 0.34%, 0.35%, 0.36%, 0.37%, 0.38%, 0.39%, 0.40%, 0.41%, 0.42%, 0.43%, 0.44%, 0.45%, 0.46%, 0.47%, 0.48%, 0.49%, 0.50%, 0.51%, 0.52%, 0.53%, 0.54%, 0.55%, 0.56%, 0.57%, 0.58%, 0.59%, 0.60%, 0.61%, 0.62%, 0.63%, 0.64%, 0.65%, 0.66%, 0.67%, 0.68%, 0.69%, 0.70%, 0.71%, 0.72%, 0.73%, 0.74%, 0.75%, 0.76%, 0.77%, 0.78%, 0.79%, 0.80%, 0.81%, 0.82%, 0.83%, 0.84%, 0.85%, 0.86%, 0.87%, 0.88%, 0.89%, 0.90%, 0.91%, 0.92%, 0.93%, 0.94%, 0.95%, 0.96%, 0.97%, 0.98%, 0.99%, 1.00%, 1.01%, 1.02%, 1.03%, 1.04%, 1.05%, 1.06%, 1.07%, 1.08%, 1.09%, 1.10%, 1.11%, 1.12%, 1.13%, 1.14%, 1.15%, 1.16%, 1.17%, 1.18%, 1.19%, or 1.20% Mn. All expressed in wt. %. As discussed herein, the aluminum alloy may include 0.30 wt. % to 1.2 wt. % Cu to compensate for the reduced content of Mg to strengthen the aluminum alloy. In some embodiments, aluminum alloys including less than 0.30 wt. % Mn may result to insufficient strength properties. In some embodiments, an aluminum alloy including greater than 0.90 wt. % Mn may lead to intermetallic phases that can deteriorate the formability and end making performance.

Magnesium (Mg)

[0066] In some examples, the aluminum alloy described herein can include Mg in an amount from 2.0% to 5.0% (e.g., from 2.2% to 5.0%, from 2.5% to 5.0%, or from 2.0 to 4.0%) based on the total weight of the alloy. For example, the alloy can include 2.0%, 2.1%, 2.2%, 2.3%, 2.4%, 2.5%, 2.6%, 2.7%, 2.8%, 2.9%, 3.0%, 3.1%, 3.2%, 3.3%, 3.4%, 3.5%, 3.6%, 3.7%, 3.8%, 3.9%, 4.0%, 4.1%, 4.2%, 4.3%, 4.4%, 4.5%, 4.6%, 4.7%, 4.8%, 4.9%, or 5.0% Mg. All expressed in wt. %. In some embodiments, aluminum alloys including less than 2.0 wt. % Mg may lead to insufficient strength properties. In some embodiments, an aluminum alloy including greater than 5.0 wt. % Mg may lead to excess strength leading to runnability issues as well as increased stress corrosion cracking and end age softening.

[0067] In some cases, the novel aluminum alloys described herein can include a Mg content that is lower than

the Mg content of a conventional AA5182 aluminum alloy and, among other elements, can include one or more of Cu or Mn in certain amounts. For example, the aluminum alloy may include at least one of Cu or Mn in the aforementioned amounts to compensate for the reduced content of Mg in the aluminum alloy. This can avoid adding additional Mg to the aluminum alloy composition, which can reduce costs.

Zinc (Zn)

[0068] In some examples, the aluminum alloy described herein includes Zn in an amount of up to 0.30% (e.g., from 0.05% to 0.25%, from 0.10% to 0.25%, or from 0.15% to 0.25%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, or 0.30% Zn. In some cases, Zn is not present in the alloy (i.e., 0%). All expressed in wt. %.

Chromium (Cr)

[0069] In some examples, the aluminum alloy described herein includes Cr in an amount of up to 0.20% (e.g., up to 0.15%, up to 0.10%, up to 0.05%, up to 0.03%, from 0.01% to 0.20%, from 0.05% to 0.20%, or from 0.05% to 0.15%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, or 0.20% Cr. In some cases, Cr is not present in the alloy (i.e., 0%). All expressed in wt. %. In some embodiments, adding Cr to the aluminum alloy composition can promote formation of alpha phase particles in the aluminum alloy microstructure. Cr addition can beneficially optimize particle size (e.g., decrease particle size) of the aluminum alloy by promoting transformation of large constituents into alpha phase particles during homogenization. The particle size and number density of particles in the aluminum alloy microstructure are important factors that determine, in part, the formability and performance of the aluminum alloy. In some embodiments, Cr can be added to of any of the aluminum alloys of Tables 1-10 to control particle size formation in the aluminum alloy microstructure.

Titanium (Ti)

[0070] In some examples, the aluminum alloy described herein includes Ti in an amount of up to 0.20% (e.g., up to 0.15%, up to 0.10%, up to 0.05%, or up to 0.03%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, or 0.20% Ti. In some cases, Ti is not present in the alloy (i.e., 0%). All expressed in wt. %.

[0071] In some examples, the aluminum alloy described herein can include a combined content of Mg, Mn, and Cu in an amount of 3.5%, 3.6%, 3.7%, 3.8%, 3.9%, 4.0%, 4.1%, 4.2%, 4.3%, 4.4%, 4.5%, 4.6%, 4.7%, 4.8%, 4.9%, or 5.0%. All expressed in wt. %. In some embodiments, an aluminum alloy including the combined wt. % of Mg, Mn, and Cu less than 3.5 wt. % may lead to insufficient buckle strength. In some embodiments, an aluminum alloy including the combined wt. % of Mg, Mn, and Cu greater than 5.0 wt. % may

lead to excess strength. In some examples, the aluminum alloy described herein can include a ratio of Mg to Cu (also referred to herein as Mg:Cu ratio) can be from 12:1 to 80:1 (e.g., from 12:1 to 70:1 to 15:1 to 70:1). For example, the Mg:Cu ratio can be 12:1, 13:1, 14:1, 15:1, 16:1, 17:1, 18:1, 19:1, 20:1, 21:1, 22:1, 23:1, 24:1, 25:1, 26:1, 27:1, 28:1, 29:1, 30:1, 31:1, 32:1, 33:1, 34:1, 35:1, 36:1, 37:1, 38:1, 39:1, 40:1, 41:1, 42:1, 43:1, 44:1, 45:1, 46:1, 47:1, 48:1, 49:1, 50:1, 51:1, 52:1, 53:1, 54:1, 55:1, 56:1, 57:1, 58:1, 59:1, 60:1, 61:1, 62:1, 63:1, 64:1, 65:1, 66:1, 67:1, 68:1, 69:1, 70:1, 71:1, 72:1, 73:1, 74:1, 75:1, 76:1, 77:1, 78:1, 79:1, or 80:1.

[0072] In some examples, the aluminum alloy described herein can include a ratio of Mn to Cu (also referred to herein as Mn:Cu ratio) can be from 2:1 to 15:1 (e.g., from 2:1 to 12:1 or from 3:1 to 12:1). For example, the Mn:Cu ratio can be 2:1, 3:1, 4:1, 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 11:1, 12:1, 13:1, 14:1, or 15:1.

Minor Elements

[0073] Optionally, the aluminum alloys described herein can further include other minor elements, sometimes referred to as impurities, in amounts of 0.05% or below, 0.04% or below, 0.03% or below, 0.02% or below, or 0.01% or below. These impurities may include, but are not limited to Sc, V, Ni, Hf, Zr, Sn, Ga, Ca, Bi, Na, Pb, or combinations thereof. Accordingly, Sc, V, Ni, Hf, Zr, Sn, Ga, Ca, Bi, Na, or Pb may be present in alloys in amounts of 0.05% or below, 0.04% or below, 0.03% or below, 0.02% or below, or 0.01% or below. The sum of all impurities does not exceed 0.15% (e.g., 0.1%). All expressed in wt. %. The remaining percentage of each alloy can be aluminum.

Aluminum Alloy Composition Based on AA3104 Alloy

[0074] The present disclosure also provides an aluminum alloy that has a similar composition to AA3104 aluminum alloy but has higher amounts of Mg. The aluminum alloy may include a Fe, Si, Cu, and Mn content similar to AA3104 aluminum alloy and may include a Mg content greater than 1.3 wt. %. This provides an aluminum alloy that can include high amounts of UBC scrap and less primary aluminum compared to AA3104 aluminum alloy, with minimal or no additional alloying elements other than Mg. These aluminum alloys can meet the minimum strength requirements for can end stock.

[0075] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 11.

TABLE 11

Element	Weight Percentage (wt. %)
Si	0.01-0.60
Fe	0.01-0.80
Cu	0.05-0.25
Mn	0.80-1.40
Mg	1.3-5.0
Cr	Up to 0.05
Zn	Up to 0.30
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0076] In some examples, an aluminum alloy as described herein can have the following elemental composition as provided in Table 12.

TABLE 12

Element	Weight Percentage (wt. %)
Si	0.10-0.50
Fe	0.20-0.70
Cu	0.10-0.25
Mn	0.80-1.0
Mg	1.5-5.0
Cr	Up to 0.05
Zn	Up to 0.25
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0077] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 13.

TABLE 13

Element	Weight Percentage (wt. %)
Si	0.22-0.32
Fe	0.50-0.65
Cu	0.16-0.20
Mn	0.80-0.92
Mg	2.0-4.0
Cr	Up to 0.03
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0078] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 14.

TABLE 14

Element	Weight Percentage (wt. %)
Si	0.25-0.32
Fe	0.45-0.55
Cu	0.16-0.20
Mn	0.80-0.92
Mg	2.0-4.0
Cr	Up to 0.03
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0079] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 15.

TABLE 15

Element	Weight Percentage (wt. %)
Si	0.25-0.35
Fe	0.45-0.55
Cu	0.16-0.20
Mn	0.80-0.92

TABLE 15-continued

Element	Weight Percentage (wt. %)
Mg	3.0-4.0
Cr	Up to 0.03
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0080] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 16.

TABLE 16

Element	Weight Percentage (wt. %)
Si	0.01-0.60
Fe	0.01-0.80
Cu	0.05-0.30
Mn	0.80-1.40
Mg	1.3-5.0
Cr	Up to 0.20
Zn	Up to 0.30
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0081] In some examples, an aluminum alloy as described herein can have the following elemental composition as provided in Table 17.

TABLE 17

Element	Weight Percentage (wt. %)
Si	0.10-0.50
Fe	0.20-0.70
Cu	0.11-0.30
Mn	0.80-1.0
Mg	1.5-5.0
Cr	Up to 0.20
Zn	Up to 0.25
Ti	Up to 0.05
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0082] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 18.

TABLE 18

Element	Weight Percentage (wt. %)
Si	0.22-0.32
Fe	0.50-0.65
Cu	0.20-0.30
Mn	0.80-0.92
Mg	2.0-4.0
Cr	Up to 0.20
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0083] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 19.

TABLE 19

Element	Weight Percentage (wt. %)
Si	0.25-0.32
Fe	0.45-0.55
Cu	0.20-0.30
Mn	0.80-0.92
Mg	2.0 -4.0
Cr	0.01-0.20
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

[0084] In some examples, the aluminum alloy can have the following elemental composition as provided in Table 20.

TABLE 20

Element	Weight Percentage (wt. %)
Si	0.25-0.35
Fe	0.45-0.55
Cu	0.16-0.30
Mn	0.80-0.92
Mg	3.0-4.0
Cr	0.05-0.15
Zn	Up to 0.25
Ti	Up to 0.03
Others	0-0.05 (each) 0-0.15 (total)
A1	

Silicon (Si)

[0085] In some examples, the aluminum alloy described herein includes Si in an amount of from 0.01% to 0.60% (e.g., from 0.10% to 0.50%, from 0.20% to 0.40%, from 0.25% to 0.35%, from 0.22% to 0.33%, or from 0.25% to 0.32%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, 0.30%, 0.31%, 0.32%, 0.33%, 0.34%, 0.35%, 0.36%, 0.37%, 0.38%, 0.39%, 0.40%, 0.41%, 0.42%, 0.43%, 0.44%, 0.45%, 0.46%, 0.47%, 0.48%, 0.49%, 0.50%, 0.51%, 0.52%, 0.53%, 0.54%, 0.55%, 0.56%, 0.57%, 0.58%, 0.59%, or 0.60% Si. All expressed in wt. %.

Iron (Fe)

[0086] In some examples, the aluminum alloy described herein also includes Fe in an amount of from 0.01% to 0.80% (e.g., from 0.05% to 0.70%, from 0.10% to 0.50%, from 0.40% to 0.60%, from 0.45% to 0.55%, from 0.50% to 0.65%, or from 0.45% to 0.55%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%,

0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, 0.30%, 0.31%, 0.32%, 0.33%, 0.34%, 0.35%, 0.36%, 0.37%, 0.38%, 0.39%, 0.40%, 0.41%, 0.42%, 0.43%, 0.44%, 0.45%, 0.46%, 0.47%, 0.48%, 0.49%, 0.5%, 0.51%, 0.52%, 0.53%, 0.54%, 0.55%, 0.56%, 0.57%, 0.58%, 0.59%, 0.60%, 0.61%, 0.62%, 0.63%, 0.64%, 0.65%, 0.66%, 0.67%, 0.68%, 0.69%, 0.70%, 0.71%, 0.72%, 0.73%, 0.74%, 0.75%, 0.76%, 0.77%, 0.78%, 0.79%, or 0.80% Fe. All expressed in wt. %.

Copper (Cu)

[0087] In some examples, the aluminum alloy described herein includes Cu in an amount of from 0.05 to 0.30% (e.g., from 0.05 to 0.25, from 0.05 to 0.25, from 0.11 to 0.30, from 0.20 to 0.30, from 0.10% to 0.25%, from 0.15% to 0.25%, or from 0.16% to 0.20%) based on the total weight of the alloy. For example, the alloy can include 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%, 0.27%, 0.28%, 0.29%, or 0.30% Cu. All expressed in wt. %.

Manganese (Mn)

[0088] In some examples, the aluminum alloy described herein can include Mn in an amount from 0.80% to 1.40% (e.g., from 0.80% to 1.20%, from 0.80% to 1.10%, or from 0.80% to 0.92%) based on the total weight of the alloy. For example, the alloy can include 0.80%, 0.81%, 0.82%, 0.83%, 0.84%, 0.85%, 0.86%, 0.87%, 0.88%, 0.89%, 0.90%, 0.91%, 0.92%, 0.93%, 0.94%, 0.95%, 0.96%, 0.97%, 0.98%, 0.99%, 1.00%, 1.01%, 1.02%, 1.03%, 1.04%, 1.05%, 1.06%, 1.07%, 1.08%, 1.09%, 1.10%, 1.11%, 1.12%, 1.13%, 1.14%, 1.15%, 1.16%, 1.17%, 1.18%, 1.19%, 1.20%, 1.21%, 1.22%, 1.23%, 1.24%, 1.25%, 1.26%, 1.27%, 1.28%, 1.29%, 1.30%, 1.31%, 1.32%, 1.33%, 1.34%, 1.35%, 1.36%, 1.37%, 1.38%, 1.39%, or 1.40% Mn. All expressed in wt. %.

Magnesium (Mg)

[0089] In some examples, the aluminum alloy described herein can include Mg in an amount from 1.3% to 5.0% (e.g., from 1.5% to 5.0%, from 2.0% to 5.0%, from 2.0% to 4.0%, from 3.0 to 4.0%, or from 1.3% to 2.0%) based on the total weight of the alloy. For example, the alloy can include 1.3%, 1.4%, 1.5%, 1.6%, 1.7%, 1.8%, 1.9%, 2.0%, 2.1%, 2.2%, 2.3%, 2.4%, 2.5%, 2.6%, 2.7%, 2.8%, 2.9%, 3.0%, 3.1%, 3.2%, 3.3%, 3.4%, 3.5%, 3.6%, 3.7%, 3.8%, 3.9%, 4.0%, 4.1%, 4.2%, 4.3%, 4.4%, 4.5%, 4.6%, 4.7%, 4.8%, 4.9% or 5.0% Mg. All expressed in wt. %. In some embodiments, the aluminum alloy described herein can include 1.3% to 2.0% Mg for applications that require less strength (e.g., can body or can end stock for water or other low pressure applications).

Zinc (Zn)

[0090] In some examples, the aluminum alloy described herein includes Zn in an amount of up to 0.30% (e.g., from 0.05% to 0.25%, from 0.10% to 0.25%, or from 0.15% to 0.25%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, 0.20%, 0.21%, 0.22%, 0.23%, 0.24%, 0.25%, 0.26%,

0.27%, 0.28%, 0.29%, or 0.30% Zn. In some cases, Zn is not present in the alloy (i.e., 0%). All expressed in wt. %.

Chromium (Cr)

[0091] In some examples, the aluminum alloy described herein includes Cr in an amount of up to 0.20% (e.g., up to 0.15%, up to 0.10%, up to 0.05%, up to 0.03%, from 0.01% to 0.20%, from 0.05% to 0.20%, or from 0.05% to 0.15%) based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.04%, 0.05%, 0.06%, 0.07%, 0.08%, 0.09%, 0.10%, 0.11%, 0.12%, 0.13%, 0.14%, 0.15%, 0.16%, 0.17%, 0.18%, 0.19%, or 0.20% Cr. In some cases, Cr is not present in the alloy (i.e., 0%). All expressed in wt. %. In some embodiments, adding Cr to the aluminum alloy composition can promote formation of alpha phase particles in the aluminum alloy microstructure. Cr addition can beneficially optimize particle size (e.g., decrease overall particle size) of the aluminum alloy by promoting transformation of large constituents into alpha phase particles during homogenization. The particle size and number density of particles in the aluminum alloy microstructure are important factors that determine, in part, the formability and performance of the aluminum alloy. In some embodiments, Cr can be added to of any of the aluminum alloys of Tables 11-20 to control particle size formation in the aluminum alloy microstructure.

[0092] In some embodiments, the amounts of Cr and Mg are carefully controlled to promote alpha phase particles. It was found that adding additional Cr and reducing the amount of Mg in the aforementioned aluminum alloy compositions in Tables 11-20 can beneficially promote formation of alpha phase particles. In some embodiments, the addition of Cr and the reduction of Mg in the aluminum alloy composition can be proportional. For example, if Cr is added in amount of 0.05 wt. %, the Mg content can be reduced by 0.50 wt. % to 1.00 wt. %. As one example, if the aluminum alloy includes 0.10 wt. % of Cr, the Mg content of the aluminum alloy can be reduced by 0.50 wt. %. In some examples, the aluminum alloy described herein can include a ratio of Mg to Cr (also referred to herein as Mg:Cr ratio) can be from 20:1 to 70:1 (e.g., from 25:1 to 65:1 or from 25:1 to 35:1). For example, the Mg:Cr ratio can be 20:1, 25:1, 30:1, 35:1, 40:1, 45:1, 50:1, 55:1, 60:1, 65:1, or 70:1. In some embodiments, including Mg and Cr in the aforementioned ratios produces an aluminum alloy having good strength and formability of the aluminum alloy.

[0093] In some examples, the aluminum alloy described herein can include a combined content of Si, Cr, and Cu in an amount of greater than 0.35% (e.g., greater than 0.36%, 0.37%, 0.38%, 0.39%, 0.40%, 0.41%, 0.42%, 0.43%, 0.44%, 0.45%, 0.46%, 0.47%, 0.48%, 0.49%, or 0.50%). All expressed in wt. %.

Titanium (Ti)

[0094] In some examples, the aluminum alloy described herein includes Ti in an amount of 0.05% or below, 0.04% or below, 0.03% or below, 0.02% or below or 0.01% or below based on the total weight of the alloy. For example, the alloy can include 0.01%, 0.02%, 0.03%, 0.040% or 0.05% Ti. In some cases, Ti is not present in the alloy (i.e., 0%). All expressed in wt. %.

Minor Elements

[0095] Optionally, the aluminum alloys described herein can further include other minor elements, sometimes referred to as impurities, in amounts of 0.05% or below, 0.04% or below, 0.03% or below, 0.02% or below, or 0.01% or below. These impurities may include, but are not limited to Sc, V, Ni, Hf, Zr, Sn, Ga, Ca, Bi, Na, Pb, or combinations thereof. Accordingly, Sc, V, Ni, Hf, Zr, Sn, Ga, Ca, Bi, Na, or Pb may be present in alloys in amounts of 0.05% or below, 0.04% or below, 0.03% or below, 0.02% or below, or 0.01% or below. The sum of all impurities does not exceed 0.15% (e.g., 0.1%). All expressed in wt. %. The remaining percentage of each alloy can be aluminum.

Recycled Content

[0096] The aluminum alloys described herein can tolerate higher amounts of recycled aluminum alloy scrap and still exhibit desirable mechanical properties. The impact of the impurities and/or alloying elements on the mechanical properties of the aluminum alloy is reduced by providing a tailored aluminum alloy composition to compensate for the impurities. This enables a higher amount of less expensive, higher impurity recycled aluminum materials (e.g., used beverage can) for producing aluminum alloys that can still exhibit desirable properties. The aluminum alloy compositions described herein can include higher amounts of recycled aluminum alloy with little or no additional primary aluminum and a reduced amount of more expensive alloying elements (e.g., Mg).

[0097] In some embodiments, the aluminum alloy composition described herein provides a composition that is well-suited for utilizing used beverage can (UBC) scrap or other aluminum alloy containers as recycle material. UBC scrap is a mixture of various aluminum alloys (e.g., from different aluminum alloys used for can bodies and can ends) and can often include foreign substances, such as rainwater, drink remainders, organic matter (e.g., paints and laminated films), and other materials. UBC scrap generally includes a mixture of metal from various aluminum alloys, such as metal from can bodies (e.g., AA3104, AA3004, or other 3xxx series aluminum alloys) and can ends (e.g., AA5182 or other 5xxx series aluminum alloys). UBC scrap can be shredded and de-coated or de-lacquered prior to being melted for use as liquid metal stock in casting a new metal product.

[0098] As discussed herein, the aluminum alloy composition described herein reduces the compositional gap between can body stock and can end stock. This allows the use of more recycled aluminum alloy, particularly UBC scrap, for producing can end stock and reduces the amount of both primary aluminum and additional alloying elements (e.g., Mg). In some aspects, the aluminum alloys described herein include a high amount of UBC scrap at or greater than 25%, e.g., at or greater than 30%, at or greater than 35%, at or greater than 40%, at or greater than 45%, at or greater than 50%, at or greater than 55%, at or greater than 60%, at or greater than 65%, at or greater than 70%, or at or greater than 75%. In terms of ranges, the aluminum alloys described herein can include from 25% to 100% UBC scrap (e.g., from 25% to 95%, from 30% to 90%, from 35% to 85%, from 40% to 80%, from 50% to 70%, or from 35% to 50%).

[0099] As discussed above, in some aspects the UBC scrap includes a mixture of alloys including a 3xxx series

aluminum alloy and a 5xxx series aluminum alloy. In some aspects, the UBC scrap can include a 5xxx series aluminum alloy in an amount from 0% to 75% (e.g., from 5% to 70%, from 10% to 65%, from 15% to 60%, from 20% to 50%, or from 25% to 40%), based on the total weight of the recycled scrap. For example, the UBC scrap can include greater than 0% of a 5xxx series aluminum alloy scrap (e.g., greater than 1%, greater than 5%, greater than 10%, greater than 15%, greater than 20%, or greater than 25%), based on the total weight of the UBC scrap. All are expressed in wt. %.

[0100] In some aspects, the UBC scrap can include a 3xxx series aluminum alloy scrap (from the mixed alloy scrap) in an amount from 0% to 75% (e.g., from 5% to 70%, from 10% to 65%, from 15% to 60%, from 20% to 50%, or from 25% to 40%), based on the total weight of the UBC scrap. For example, the UBC scrap can include greater than 0% of a 3xxx series aluminum alloy scrap (e.g., greater than 1% greater than 5%, greater than 10%, greater than 15%, greater than 20%, or greater than 25%), based on the total weight of the UBC scrap. All are expressed in wt. %.

[0101] In some aspects, the aluminum alloys described herein include less than 35% primary aluminum, e.g., less than 34%, less than 33%, less than 32%, less than 31%, less than 30%, less than 29%, less than 28%, less than 27%, less than 26%, less than 25%, less than 24%, less than 23%, less than 22%, less than 21%, or less than 20%. All are expressed in wt. %.

Properties

[0102] In some examples, an aluminum alloy product (e.g., an aluminum alloy sheet) produced from the aluminum alloys described herein can have a yield strength of about 260 MPa or greater. For example, an aluminum alloy product produced from the aluminum alloys described herein can have a yield strength of 270 MPa or greater, 280 MPa or greater, 290 MPa or greater, 300 MPa or greater, 310 MPa or greater, 320 MPa or greater, 325 MPa or greater, 330 MPa or greater, 335 MPa or greater, 340 MPa or greater, 345 MPa or greater, 350 MPa or greater, 355 MPa or greater, 360 MPa or greater, 365 MPa or greater, or 370 MPa or greater. In some cases, the yield strength is from about 260 MPa to about 420 MPa (e.g., from about 280 MPa to about 450 MPa, from about 300 MPa to about 425 MPa, or from about 325 MPa to about 400 MPa), or anywhere in between. The aluminum alloy products described herein can exhibit the yield strengths as described herein when measured in a longitudinal (L) direction, a transverse (T) direction, and/or in a diagonal (D) direction, each respective to the rolling direction.

[0103] In some examples, an aluminum alloy product produced from the aluminum alloys described herein can have an ultimate tensile strength of about 280 MPa or greater. For example, the aluminum alloy products can have an ultimate tensile strength of 290 MPa or greater, 300 MPa or greater, 310 MPa or greater, 320 MPa or greater, 330 MPa or greater, 340 MPa or greater, 350 MPa or greater, 355 MPa or greater, 360 MPa or greater, 365 MPa or greater, 370 MPa or greater, 375 MPa or greater, 380 MPa or greater, 385 MPa or greater, 390 MPa or greater, 395 MPa or greater, or 400 MPa or greater. In some cases, the ultimate tensile strength is from about 280 MPa to about 550 MPa (e.g., from about 300 MPa to about 500 MPa, from about 350 MPa to about 475 MPa, or from about 375 MPa to about 430 MPa), or anywhere in between. The aluminum alloy products

described herein can exhibit the ultimate tensile strengths as described herein when measured in a longitudinal (L) direction, a transverse (T) direction, and/or in a diagonal (D) direction, each respective to the rolling direction.

[0104] In some examples, an aluminum alloy product produced from the aluminum alloys described herein can have a total elongation from about 4% to 20% (e.g., from 4% to 18%, from 5% to 16%, from 5.5% to 14%, from 6% to 12%, or from 6.5% to 10%). For example, an aluminum alloy product produced from the aluminum alloys described herein can have a total elongation of about 4%, 5%, 6%, 7%, 8%, 9%, 10%, 11%, 12%, 13%, 14%, 15%, 16%, 17%, 18%, 19%, or 20%, or anywhere in between. The aluminum alloy products described herein can exhibit the total elongations as described herein when measured in a longitudinal (L) direction, a transverse (T) direction, and/or in a diagonal (D) direction, each respective to the rolling direction.

Methods of Making Aluminum Alloys

Casting

[0105] The aluminum alloys described herein can be cast into a cast product using a direct chill (DC) process or can be cast using a continuous casting (CC) process. The casting process is performed according to standards commonly used in the aluminum industry as known to one of skill in the art. The CC process may include, but is not limited to, the use of twin belt casters, twin roll casters, or block casters. In some examples, the casting process is performed by a CC process to form a slab, a strip, or the like. In some examples, the casting process is a DC casting process to form a cast product.

[0106] The cast product, slab, or strip can then be subjected to further processing steps. Optionally, the further processing steps can be used to prepare aluminum alloy products (e.g., sheets, shales, or plates). Such processing steps include, but are not limited to, a homogenization step, a hot rolling step, a cold rolling step, and an optional lacquering step. The processing steps are described below in relation to a cast product. However, the processing steps can also be used for a cast slab or strip, using modifications as known to those of skill in the art.

Homogenization Practice 1

[0107] The homogenization conditions can optimize particle formation (e.g., produce higher amounts of alpha phase particles that can be broken down into small particles) in the aluminum alloy resulting in good mechanical properties. The particle size and particle number density determine the formability and performance of the aluminum alloy. In some embodiments, the particle size in the aluminum alloy microstructure can be controlled by modifying the alloying elements (e.g., adding Cr and reducing Mg) which can promote the transformation of constituent particles from large particles (e.g., $Al_x(Fe,Mn)$) to alpha phase particles (e.g., $Al-(Fe,Mn)Si$) during homogenization. Therefore, homogenization practice in combination with the aluminum alloy composition can promote particle breakdown and small particle sizes in the aluminum alloy microstructure.

[0108] Aluminum alloys produced from a high content of recycled aluminum alloy materials include high amounts Fe, Si, and/or Mn compared to aluminum alloys that do not include a high content of recycled aluminum alloy materials.

For example, aluminum alloys produced from used beverage cans or other aluminum alloy scraps include high amounts of Fe, Si, and/or Mn. These additional alloying elements result in a greater number density of large particles that are difficult to breakdown. Large particles, for example, $Al_x(Fe,Mn)$ particles, are difficult to breakdown while alpha phase particles can be easily broken into smaller particles during the rolling processes. Therefore, it is beneficial to transform $Al_x(Fe,Mn)$ to alpha phase particles for good mechanical properties of the aluminum alloy.

[0109] FIG. 13A shows a scanning electron microscopy (SEM) image of the microstructure of a AA5182 aluminum alloy and FIG. 1B shows the SEM image of the microstructure for a recycle friendly aluminum alloy including 0.40 wt. % Fe, 0.16 wt. % Si, 0.60 wt. % Mn, 0.07 wt. % Cu, 4.0 wt. % Mg, and remainder Al. Both alloys are used to produce aluminum alloys for can end stock. The recycle friendly aluminum alloy is produced from a high content of recycled aluminum alloy materials, whereas AA5182 aluminum alloy has a strictly controlled aluminum alloy composition and is produced from very little recycled aluminum alloy materials. Due to the higher recycled content of the recycle friendly aluminum alloy, higher amounts of alloying elements are present in the aluminum alloy. For example, the recycle friendly aluminum alloy includes higher amounts of Si, Mn, and Fe than the AA5182 aluminum alloy. As shown in FIG. 13B, the higher amounts of alloying elements present in the recycle friendly aluminum alloy leads to formation of particles having large sizes (e.g., larger than 5 microns) in the aluminum alloy microstructure. In comparison, FIG. 13A shows that the microstructure of AA5182 aluminum alloy includes particles having smaller sizes and a lower number density of large particles. The larger constituents negatively affect the properties of the aluminum alloy. For example, the larger constituents result in poor formability. It was surprisingly found that the particle size in the aluminum alloy microstructure can be controlled by modifying the alloying elements (e.g., adding Cr and reducing Mg) in the aluminum alloy composition and homogenizing the aluminum under specific conditions to promote the transformation of constituent particles from large particles to alpha phase particles. For example, altering the aluminum alloy composition to lower the amount of Mg and adding higher levels of Cr, in combination with homogenization, can aid in transforming larger constituents in the aluminum alloy microstructure into smaller alpha phase particles. For example, the method described herein may transform $Al_x(Fe,Mn)$ constituents into alpha phase particles during homogenization. In some embodiments, the homogenization conditions can aid in transforming large particles into alpha phase particles.

[0110] In some embodiments, a cast product may be heated to a homogenization temperature ranging from about 450° C. to about 570° C. For example, the cast product can be heated to a temperature of 450° C., 460° C., 470° C., 480° C., 490° C., 500° C., 510° C., 520° C., 530° C., 540° C., 550° C., 560° C., or 570° C. In some embodiments, the cast product can be homogenized at a homogenization temperature range from 450° C. to 570° C. (e.g., from 500° C. to 560° C. or 525° C. to 550° C.) produces the highest concentration of stable alpha phase particles. As discussed above, the aluminum alloys described herein provide a larger homogenization temperature window to produce alpha phase particles. Specifically, homogenizing the alu-

minum alloys in Tables 6-10 and 16-20 in the homogenization temperature window from 450° C. to 570° C. can transform large constituents to alpha phase particles in the aluminum alloy microstructure across the entire homogenization temperature range. This beneficially provides increased amounts of transformation of large constituents to alpha phase particles during homogenization.

[0111] In some embodiments, the heating rate to the homogenization temperature can be about 70° C./hour or less, about 60° C./hour or less, or about 50° C./hour or less. The cast product may then be allowed to soak (i.e., held at the indicated temperature) for a period of time at the homogenization temperature to form a homogenized product. In some examples, the total time for the homogenization step, including the heating and soaking phases, can be up to about 10 hours. For example, the cast product can be soaked at the homogenization temperature for 10 minutes, 30 minutes, 1 hour, 2 hours, 3 hours, 4 hours, 5 hours, 6 hours, 7 hours, 8 hours, 9 hours, or 10 hours.

Homogenization Practice 2

[0112] In a homogenization step, a cast product may be heated to a homogenization temperature, such as a temperature ranging from about 400° C. to about 600° C. For example, the cast product can be heated to a temperature of 400° C., 410° C., 420° C., 430° C., 440° C., 450° C., 460° C., 470° C., 480° C., 490° C., 500° C., 510° C., 520° C., 530° C., 540° C., 550° C., 560° C., 570° C., 580° C., 590° C., or 600° C. In some embodiments, the heating rate to the peak metal temperature can be about 70° C./hour or less, about 60° C./hour or less, or about 50° C./hour or less. The cast product may then be allowed to soak (i.e., held at the indicated temperature) for a period of time to form a homogenized product. In some examples, the total time for the homogenization step, including the heating and soaking phases, can be up to about 10 hours.

[0113] In some embodiments, the homogenization step described herein can be a two-stage homogenization. The first stage may include heating a cast product to a first homogenization temperature of about 350° C. to about 450° C. (e.g., from 360° C. to 440° C., from 370° C. to 430° C., from 380° C. to 420° C., or from 400° C. to 420° C.). For example, the cast product can be heated to a temperature of about 350° C., 360° C., 370° C., 380° C., 390° C., 400° C., 410° C., 420° C., 430° C., 440° C., or 450° C. In some cases, the cast product is heated to a first homogenization temperature from 400° C. to 420° C. In some cases, the heating rate to the first homogenization temperature can be about 70° C./hour or less, about 60° C./hour or less, or about 50° C./hour or less. The cast product is then allowed to soak (i.e., held at the indicated temperature first homogenization temperature) for a period of time. In some cases, the cast product is allowed to soak for up to 5 hours (e.g., from 30 minutes to 5 hours, inclusively) at the first homogenization temperature. For example, the cast product can be soaked at a first homogenization temperature from 350° C. to 450° C. for 30 minutes, 1 hour, 2 hours, 3 hours, 4 hours, or 5 hours. In some embodiments, the cast product can be soaked at a first homogenization temperature from 400° C. to 420° C. for 1 hour to 2 hours.

[0114] The second stage may include heating the cast product from the first homogenization temperature to a second homogenization temperature of about 450° C. to about 550° C. (e.g., from 460° C. to 540° C., from 470° C.

to 530° C., from 480° C. to 520° C., or from 480° C. to 510° C.). For example, the cast product can be heated to a temperature of about 450° C., 460° C., 470° C., 480° C., 490° C., 500° C., 510° C., 520° C., 530° C., 540° C., or 550° C. In some cases, the cast product is heated to a first homogenization temperature from 480° C. to 510° C. The cast product is then allowed to soak for a period of time at the second homogenization temperature. In some cases, the cast product is allowed to soak for up to 5 hours (e.g., from 30 minutes to 5 hours, inclusively) at the second homogenization temperature. For example, the cast product can be soaked at a second homogenization temperature from 450° C. to 550° C. for 30 minutes, 1 hour, 2 hours, 3 hours, 4 hours, or 5 hours. In some embodiments, the cast product can be soaked at a second homogenization temperature from 480° C. to 510° C. for 0 to 2 hours.

Hot Rolling

[0115] Following a homogenization step, a hot rolling step can be performed. The homogenized product can be hot rolled using a rolling mill to produce a hot rolled product. Prior to the start of hot rolling, the homogenized product can be allowed to cool to a desired temperature, such as from about 200° C. to about 425° C. For example, the homogenized product can be allowed to cool to a temperature of from about 200° C. to about 400° C., about 250° C. to about 375° C., about 300° C. to about 425° C., or from about 350° C. to about 400° C. The homogenized product can then be hot rolled at a hot rolling temperature, for example, from about 200° C. to about 450° C., to produce a hot rolled product (e.g., a hot rolled plate, a hot rolled shate, or a hot rolled sheet).

Cold Rolling

[0116] The hot rolled product can be cold rolled using cold rolling mills into thinner products, such as a final gauge rolled product. The final gauge rolled product can have a gauge between about 0.5 to about 10 mm, e.g., between about 0.7 to about 6.5 mm. Optionally, the final gauge rolled product can have a gauge of about 0.5 mm, about 1.0 mm, about 1.5 mm, about 2.0 mm, about 2.5 mm, about 3.0 mm, about 3.5 mm, about 4.0 mm, about 4.5 mm, about 5.0 mm, about 5.5 mm, about 6.0 mm, about 6.5 mm, about 7.0 mm, about 7.5 mm, about 8.0 mm, about 8.5 mm, about 9.0 mm, about 9.5 mm, or about 10.0 mm. The cold rolling can be performed to result in a final gauge thickness that represents a gauge reduction of up to about 85% (e.g., up to about 10%, up to about 20%, up to about 30%, up to about 40%, up to about 50%, up to about 60%, up to about 70%, up to about 80%, or up to about 85% reduction) as compared to a gauge prior to the start of cold rolling. In some embodiments, the cold rolling step may include one or more cold rolling steps to achieve the desired gauge thickness reduction. Optionally, the process for producing the aluminum alloy can include an interannealing step (e.g., between one or more cold rolling steps).

Lacquering

[0117] Subsequently, final gauge rolled product can optionally undergo a lacquering step. The lacquering step can apply a coating on the final gauge rolled product at a temperature from 150° C. to 400° C. for 1 second to 10 minutes. For example, the final gauge rolled product can be

lacquered at a temperature of from 150° C. to 400° C., from 200° C. to 400° C., from 250° C. to 350° C., from 200° C. to 300° C., or from 300° C. to 400° C. The peak metal temperature of the final gauge rolled product during the lacquering process may range from 100° C. to 300° C. (e.g., from 125° C. to 275° C., from 150° C. to 250° C., or from 200° C. to 300° C.).

Aluminum Alloy Microstructure

[0118] The aluminum alloys described herein include small particles in the aluminum alloy microstructure. A substantial amount of the particles present in the aluminum alloy microstructure have a particle size, measured by area, of 1.45 μm^2 or less. For example, the particle size, as measured by area, can be 1.40 μm^2 or less, 1.35 μm^2 or less, 1.30 μm^2 or less, 1.25 μm^2 or less, 1.20 μm^2 or less, 1.15 μm^2 or less, or 1.10 μm^2 or less. In some examples, the particle size ranges from 0.80 μm^2 to 1.45 μm^2 (e.g., from 0.85 μm^2 to 1.40 μm^2 or from 0.90 μm^2 to 1.35 μm^2). As used herein, a “substantial amount” as related to the number of particles represents at least 50% of the particles present in the aluminum alloy. For example, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 85%, at least 90%, at least 95%, or at least 99% of the particles present in the aluminum alloy products have a particle size of 1.45 μm^2 or less.

[0119] The aluminum alloys described herein include alpha phase particles. In some embodiments, the aluminum alloy alloys described herein include a particle area % of alpha phase particles, as measured by volume, of 1.5% or greater, 1.6% or greater, 1.7% or greater, 1.8% or greater, 1.9% or greater, or 2.0% or greater. In some embodiments, the aluminum alloy alloys described herein include a particle area % of alpha phase particles, as measured by volume, from 1.5% to 2.5% (e.g., from 1.5% to 2.0%, from 2.0% to 2.5%, or from 1.7% to 2.1%).

[0120] The aluminum alloys described herein include a ratio of alpha phase particles to Alx(Fe,Mn) that beneficially provides good formability properties for the aluminum alloy. In some embodiments, the ratio of alpha phase particles to Alx(Fe,Mn) in the aluminum alloy microstructure is from 5:1 to 40:1 (e.g., from 6:1 to 30:1, from 8:1 to 25:1, from 8:1 to 20:1, or from 10:1 to 20:1). In some embodiments, the ratio of alpha phase particles to Alx(Fe,Mn) in the aluminum alloy microstructure is 5:1, 6:1, 7:1, 8:1, 9:1, 10:1, 11:1, 12:1, 13:1, 14:1, 15:1, 16:1, 17:1, 18:1, 19:1, 20:1, 21:1, 22:1, 23:1, 24:1, 25:1, 26:1, 27:1, 28:1, 29:1, 30:1, 31:1, 32:1, 33:1, 34:1, 35:1, 36:1, 37:1, 38:1, 39:1, or 40:1.

Illustrations of Suitable Methods and Alloy Products

[0121] Illustration 1 is an aluminum alloy comprising 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

[0122] Illustration 2 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.10-0.30 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-1.0 wt. % Mn, 2.2-5.0 wt. % Mg, up to 0.15 wt. % Cr, up to 0.30 wt. % Zn, up to 0.15 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

[0123] Illustration 3 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy

comprises 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.10 wt. % Cr, up to 0.25 wt. % Zn, up to 0.10 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

[0124] Illustration 4 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.20-0.35 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.05 wt. % Cr, up to 0.25 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

[0125] Illustration 5 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.20-0.35 wt. % Si, 0.40-0.60 wt. % Fe, 0.15-0.25 wt. % Cu, 0.60-1.2 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.03 wt. % Cr, up to 0.20 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % impurities, and Al.

[0126] Illustration 6 is the aluminum alloy of any preceding or subsequent illustration, wherein a ratio of Mg:Cu is from 10:1 to 80:1 and wherein a ratio of Mn:Cu is from 2:1 to 15:1.

[0127] Illustration 7 is the aluminum alloy of any preceding or subsequent illustration, wherein a ratio of Mg:Cu is from 15:1 to 70:1 and wherein a ratio of Mn:Cu is from 3:1 to 12:1.

[0128] Illustration 8 is the aluminum alloy of any preceding or subsequent illustration, wherein a combined content of Fe and Si is greater than 0.40 wt. %.

[0129] Illustration 9 is the aluminum alloy of any preceding or subsequent illustration, wherein a combined content of Mg, Mn, and Cu is from 3.5 wt. % to 5.0 wt. %.

[0130] Illustration 10 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises at least 40 wt. % of recycled scrap.

[0131] Illustration 11 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises less than 30 wt. % of primary aluminum.

[0132] Illustration 12 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has propagation energy of at least 30.0 KJ/m² as measured by ASTM B871-1 (2021).

[0133] Illustration 13 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has a yield strength of at least 340 MPa.

[0134] Illustration 14 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has an ultimate tensile strength of at least 380 MPa.

[0135] Illustration 15 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has a total elongation of at least 4%.

[0136] Illustration 16 is a can end stock comprising the aluminum alloy of any preceding or subsequent illustration.

[0137] Illustration 17 is a method of producing an aluminum alloy, comprising: casting an aluminum alloy to form a cast product, wherein the aluminum alloy comprises 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al; homogenizing the cast product; hot rolling the cast product to produce a hot rolled product; cold rolling the hot rolled product to produce a final gauge rolled product; and optionally annealing the final gauge rolled product.

[0138] Illustration 18 is the method of any preceding or subsequent illustration, wherein the method may further comprise lacquering and curing the final gauge rolled product.

[0139] Illustration 19 is the method of any preceding or subsequent illustration, wherein the homogenization step comprises a first homogenization step and a second homogenization step.

[0140] Illustration 20 is the method of any preceding or subsequent illustration, wherein the first homogenization step comprises soaking the cast product at a temperature from 375° C. to 450° C. for 0.5 hours to 5 hours.

[0141] Illustration 21 is the method of any preceding or subsequent illustration, wherein the second homogenization step comprises soaking the cast product at a temperature from 450° C. to 550° C. for 0.01 hours to 5 hours.

[0142] Illustration 22 is the method of any preceding or subsequent illustration, wherein the aluminum alloy includes 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.15 wt. % Cr, up to 0.25 wt. % Zn, up to 0.15 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

[0143] Illustration 23 is the method of any preceding or subsequent illustration, wherein a ratio of Mg:Cu is from 15:1 to 70:1 and wherein a ratio of Mn:Cu is from 3:1 to 12:1.

[0144] Illustration 24 is a metal product prepared from the method of any preceding or subsequent illustration.

[0145] Illustration 25 is a metal product prepared from the method of any preceding or subsequent illustration, wherein the metal product is a can end stock.

[0146] Illustration 26 is an aluminum alloy including 0.01-0.60 wt. % Si, 0.01-0.80 wt. % Fe, 0.05-0.30 wt. % Cu, 0.80-1.40 wt. % Mn, 1.3-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0147] Illustration 27 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.10-0.50 wt. % Si, 0.20-0.70 wt. % Fe, 0.11-0.30 wt. % Cu, 0.80-1.00 wt. % Mn, 1.5-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0148] Illustration 28 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.22-0.32 wt. % Si, 0.50-0.65 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0149] Illustration 29 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.25-0.32 wt. % Si, 0.45-0.55 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, 0.01-0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0150] Illustration 30 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.25-0.35 wt. % Si, 0.45-0.55 wt. % Fe, 0.16-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 3.0-4.0 wt. % Mg, 0.05-0.15 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0151] Illustration 31 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.05-0.20 wt. % Cr.

[0152] Illustration 32 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has a ratio of Mg:Cr is from 20:1 to 70:1.

[0153] Illustration 33 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy has a combined content of Si, Cr, and Cu, is greater than 0.35 wt. %.

[0154] Illustration 34 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises a combined content of Fe and Si is greater than 0.40 wt. %.

[0155] Illustration 35 is the aluminum alloy of any preceding or subsequent illustration, wherein the particles in the aluminum alloy microstructure have a particle size, measured by area, of 145 μm^2 or less.

[0156] Illustration 36 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises at least 40 wt. % of recycled scrap.

[0157] Illustration 37 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises less than 30 wt. % of primary aluminum. In some embodiments, the aluminum alloy has a yield strength of at least 340 MPa.

[0158] Illustration 38 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises an ultimate tensile strength of at least 380 MPa.

[0159] Illustration 39 is the aluminum alloy of any preceding or subsequent illustration, wherein the aluminum alloy comprises a total elongation of at least 4%.

[0160] Illustration 40 is a can end stock comprising the aluminum alloy of any preceding or subsequent illustration.

[0161] Illustration 41 is a method for producing an aluminum alloy is provided comprising casting an aluminum alloy to form a cast product, wherein the aluminum alloy comprises 0.01-0.60 wt. % Si, 0.01-0.80 wt. % Fe, 0.05-0.30 wt. % Cu, 0.80-1.40 wt. % Mn, 1.3-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al, homogenizing the cast product, wherein homogenizing the cast product produces alpha phase particles; hot rolling the cast product to produce a hot rolled product; cold rolling the hot rolled product to produce a final gauge rolled product; and optionally annealing the final gauge rolled product.

[0162] Illustration 42 is the method of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.05-0.20 wt. % Cr, and wherein a ratio of Mg:Cr is from 20:1 to 70:1.

[0163] Illustration 43 is the method of any preceding or subsequent illustration, wherein the homogenization step comprises heating and soaking the cast product at a temperature from 450° C. to 570° C., wherein homogenization is configured to transform large particles to alpha phase particles.

[0164] Illustration 44 is the method of any preceding or subsequent illustration, wherein the cast product is soaked at the homogenization temperature for up to 10 hours.

[0165] Illustration 45 is the method of any preceding or subsequent illustration, wherein the aluminum alloy includes a particle area % of alpha phase particles, as measured by volume, of 1.5% or greater after homogenization.

[0166] Illustration 46 is the method of any preceding or subsequent illustration, wherein the aluminum alloy comprises 0.22-0.32 wt. % Si, 0.50-0.65 wt. % Fe, 0.20-0.30 wt. % Cu, 0.80-0.92 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.25 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % of impurities, and remainder Al.

[0167] The following examples will serve to further illustrate the present invention without, however, constituting any limitation thereof. On the contrary, it is to be clearly understood that resort may be had to various embodiments, modifications, and equivalents thereof which, after reading the description herein, may suggest themselves to those skilled in the art without departing from the spirit of the invention.

[0168] During the studies described in the following examples, conventional procedures were followed, unless otherwise stated. Some of the procedures are described below for illustrative purposes.

EXAMPLES

Example 1

[0169] Sample aluminum alloys were tested to determine the properties of the aluminum alloys described herein. Comparative Example 1 and Examples 1-5 were prepared according to the methods described herein. Comparative Example 1 was prepared from a conventional AA5182 aluminum alloy, which is currently employed as can end stock. Examples 1-5 were prepared from aluminum alloys described herein. Table 21 provides the aluminum alloy composition for each of Comparative Example 1 and Examples 1-5.

TABLE 21

	Si (wt. %)	Fe (wt. %)	Cu (wt. %)	Mn (wt. %)	Mg (wt. %)	Impurities (wt. %)	Al	Primary (wt. %)
Comp. 1	0.14	0.34	0.07	0.33	4.9	Up to 0.15	Bal.	35
Ex. 1	0.16	0.40	0.07	0.33	4.9	Up to 0.15	Bal.	25-30
Ex. 2	0.16	0.40	0.07	0.60	4.9	Up to 0.15	Bal.	25-30
Ex. 3	0.16	0.40	0.07	0.60	4.1	Up to 0.15	Bal.	25-30
Ex. 4	0.16	0.40	0.07	0.80	3.5	Up to 0.15	Bal.	25-30
Ex. 5	0.18	0.45	0.07	0.80	3.5	Up to 0.15	Bal.	20
Ex. 6	0.28	0.53	0.18	0.86	3.3	Up to 0.15	Bal.	<10

[0170] As shown in Table 21, Comparative Example 1 includes lower amounts of Si and Fe compared to Examples 1-5. Although Comparative Example 1 and Example 1 have similar compositions, the difference of Fe and Si is substantial for aluminum alloys used to produce can end stock and may cause significant differences in performance. Comparative Example 1 includes 35 wt. % primary aluminum. Examples 1-4 include from 25% to 30 wt. % primary aluminum and Example 5 includes 20 wt. % primary aluminum. The aluminum alloys described herein include up to 15 wt. % less primary aluminum than AA5182 aluminum alloy, which can be a substantial cost savings. Additionally, Examples 1-5 can incorporate higher amounts of UBC scrap in place of primary aluminum because the aluminum alloys can tolerate higher amounts of Si and Fe.

[0171] FIG. 1 provides a graph of the yield strength, ultimate tensile strength, and total elongation of Comparative Example 1 and Examples 1-5. Examples 1-5 exhibited similar yield strength, ultimate tensile strength, and total elongation properties as Comparative Example 1 despite having less primary aluminum and higher UBC scrap. In fact, Examples 1-3 exhibited the same or greater yield strength and ultimate tensile strength properties as Comparative Example 1. Examples 1 and 2 also exhibited the same or better total elongation properties compared to Comparative Example 1. Examples 4 and 5 included significantly less Mg than Comparative Example 1 and achieved similar yield strength and ultimate tensile strength properties as Comparative Example 1. The additional Mn in Examples 4 and 5 may have contributed to additional strengthening that compensates for the reduced amount of Mg.

[0172] FIG. 2 provides a graph of the yield stress (MPa) when measured in a longitudinal (L) direction, a transverse (T) direction, and in a diagonal (D) direction, each respective to the rolling direction, of Comparative Example 1 and Examples 1-5. Comparative Example 1 demonstrated a yield stress of about 350 MPa in each of the L, T, and D directions. Examples 2 and 3 demonstrated a higher yield strength in the L, T, and D directions compared to Comparative Example 1 despite having a higher Si and Fe content. Example 3, having a Mn content of 0.60 wt. %, exhibited the highest yield stress in the L, T, and D directions. Examples 3 and 4, each having a Mg of 4.1 wt. %, exhibited comparable yield stress values as Comparative Example 1 despite having less Mg. Example 4 demonstrates that reducing the Mg content

of the alloy composition while increasing the Mn content can achieve a comparable yield stress in the L, T, and D directions when compared to Comparative Example 1. Additionally, Example 5 demonstrates that increasing the Fe and Si content to increase the recycle content of the alloy composition led to a yield stress that was also similar to Comparative Example 1. These results indicate that by increasing the recycled content and replacing Mg with higher levels of Mn, the resulting properties are consistent with current commercially available aluminum alloys for can end stock.

[0173] Comparative Example 1 and Examples 1, 4, and 5 were evaluated to determine the effect of Fe and Si on the particle number density and particle area % of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the alu-

minum alloy microstructure. The particle number density and particle area % results for Comparative Example 1 and Examples 1, 4, and 5 are shown in FIGS. 3A and 3B. Example 1, which included higher amounts of Si and Fe than Comparative Example 1, had an overall particle area percent of greater than 1.50%, which is greater than Comparative Example 1. Example 1 exhibited a particle size that was smaller than Comparative Example 1 despite having a higher number density of alpha, Al(Fe,Mn), and Mg₂Si particles. Examples 4 and 5, which both had a lower Mg content and higher Mn content than Comparative Example 1, exhibited a greater particle size alpha area % than Comparative Example 1. As discussed herein, increasing the Si and Fe content in the aluminum alloy composition allows for higher recycle content. For example, an aluminum alloy composition with increased Si and Fe (e.g., Example 5) maintained similar particle size to Example 4 while the particle area of Al(Fe,Mn) increased. Additionally, increasing the content of Fe, Si and Mn, while simultaneously decreasing the Mg content (Examples 4 and 5) led to a comparable overall particle number density to Comparative Example 1 while significantly increasing the alpha particle number density and decreasing the particle number density of Al(Fe,Mn) and Mg₂Si.

[0174] Comparative Example 1 and Examples 1, 4, and 5 samples were formed into 1 mm thick rectangular samples for crack propagation testing according to the Kahn tear test (ASTM B871-10 (2021)). The samples were tested for fracture toughness via crack propagation energy using a sample with a pre-existing crack while a cyclical load is applied to each side of the crack allowing it to grow. FIG. 4 shows the results of the Kahn tear test for Comparative Example 1 and Examples 1, 4, and 5. Example 1 had a lower propagation energy and a higher yield stress relative to Comparative Example 1. Additionally, Example 4, which had a lower Mg content and higher Si and Fe content than Comparative Example 1, exhibited a lower propagation energy while maintaining a relatively similar yield stress when compared to Comparative Example 1. The results indicate that the aluminum alloys described herein, including lower amounts of Mg and higher amounts of Si and Fe, exhibit similar propagation energy and yield stress when compared to Comparative Example 1. Moreover, the increased amounts of Si, Fe and Mn combined with lower levels of Mg further increase the recycled content of the aluminum alloys while still providing comparable properties to current AA5182 aluminum alloys.

[0175] Examples 1 and 2 were evaluated to determine the effect of Mn on the particle number density and particle area % of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the aluminum alloy microstructure. FIGS. 5A and 5B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of Examples 1 and 2. Example 1 included 0.16 wt. % Si, 0.40 wt. % Fe, and 0.33 wt. % Mn and Example 2 included the same amounts of Si and Fe, and a higher Mn content of 0.60 wt. %. Example 2 exhibited a higher particle size and particle area of alpha

particles and Al(Fe,Mn) particles than Example 1. The higher amount of Mn in Example 2 also increased the particle number density compared to Example 1.

[0176] FIG. 6 shows the results of the Kahn tear test (ASTM B871-10 (2021)) for Examples 1 and 2. Example 1 had a propagation energy of 38.8 KJ/m² and a yield stress of about 360 MPa and Example 2 had a propagation energy of 34.4 KJ/m² and a yield stress of about 375 MPa. Example 2, which had a higher Mn content than Example 1, exhibited a lower propagation energy and higher yield stress than Example 1.

[0177] Examples 2 and 3 were evaluated to determine the effect of Mg on the particle number density and particle area % of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the aluminum alloy microstructure. FIGS. 7A and 7B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of Examples 2 and 3. Example 3 included a lower Mg content (4.1 wt. %) compared to Example 2 (4.9 wt. %). Example 3 had a particle size of about 2.4 μm² and a particle area of about 2%, which was consistent with Comparative Example 1. Example 3 had a smaller particle size and a smaller area fraction of Al(Fe, Mn) compared to Example 2. By changing the Mg content from 4.9 wt. % (Example 2) to 4.1 wt. % (Example 3) the particle number density of alpha particles increased while the particle number density of both Al(Fe,Mn) and Mg₂Si decreased.

[0178] FIG. 8 shows the results of the Kahn tear test (ASTM B871-10 (2021)) for Examples 2 and 3. Alloy Example 2 had a propagation energy of 34.4 KJ/m² and a yield stress of about 375 MPa. Example 3, which had a lower Mg content than Example 2, had a propagation energy of 36.1 KJ/m² and a yield stress of about 360 MPa.

[0179] Examples 1, 3, and 4 were evaluated to determine the effect of Mg and Mn on the particle number density and particle area % of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the aluminum alloy microstructure. FIGS. 9A and 9B provide graphs of the particle area % and the particle number density of alpha particles, Al(Fe,Mn) particles, and Mg₂Si particles in the microstructure of Examples 1, 3, and 4. Example 3, which had a lower Mg and higher Mn content than Example 1, had a higher alpha particle area fraction and a lower Al(Fe,Mn) and Mg₂Si particle area fraction than Example 1. Additionally, the particle size for Examples 3 and 4 was larger relative to Example 1 as the Mg content decreased. Further, it was found that by replacing Mg with Mn the total particle area is almost consistent for each of Examples 1, 3, and 4; however, Examples 3 and 4 exhibit increase alpha area % and lower Mg₂Si area % compared to Example 1. Although the particle number density is lower for Examples 3 and 4 relative to Example, the particle size for Examples 3 and 4 were larger.

[0180] FIG. 10 shows the results of the Kahn tear test (ASTM B871-10 (2021)) for Examples 1, 3, and 4. Example 1 had a propagation energy of 38.8 KJ/m² and a yield stress of about 360 MPa. Examples 3 and 4, which had less Mg and

higher Mn than Example 1, exhibited similar properties for propagation energy and yield stress. Specifically, Example 3 had a propagation energy of 36.1 KJ/m² and a yield stress of about 355 MPa. Example 4 had a propagation energy of 32.2 KJ/m² and a yield stress of about 340 MPa. The results indicate that the aluminum alloys described herein, includ-

composition as AA3104 aluminum alloy but a lower Mg content. Examples 7 and 8 are alloys comprising a similar Fe, Si, Cu, and Mn content as AA3104 aluminum alloy with higher amounts of Mg. Table 22 provides the aluminum alloy composition for each of Comparative Examples 2 and 3 and Examples 7 and 8.

TABLE 22

	Si (wt. %)	Fe (wt. %)	Cu (wt. %)	Mn (wt. %)	Mg (wt. %)	Impurities (wt. %)	Al	Primary (wt. %)
Comp. 2	≤0.60	≤0.80	0.05-0.25	0.80-1.4	0.80-1.3	Up to 0.15	Bal.	35
Comp. 3	0.28	0.53	0.18	0.86	0	Up to 0.15	Bal.	<10
Ex. 7	0.28	0.53	0.18	0.86	2.0	Up to 0.15	Bal.	<10
Ex. 8	0.28	0.53	0.18	0.86	3.0	Up to 0.15	Bal.	<10

ing lower amounts of Mg and higher amounts of Mn, exhibit similar propagation energy. Moreover, the results suggest that the alloy composition can compensate lower Mg content with an increase of Mn, Si and Fe to increase the recycled content while still maintaining comparable physical properties as AA5182 aluminum alloy.

[0181] Comparative Example 1 and Examples 1-5 were formed into 1 mm thick disks to test formability using the mini bulge test according to ISO 16808 (2022). FIG. 11 shows the results of the mini-bulge test for Comparative Example 1 and Examples 1-5. For example, Comparative example 1 had an average strain to failure of 0.170 while Example 1, which had a higher Fe and Si content, had a relatively comparable average strain to failure of 0.168. Further, Example 2, which had a higher Mn content than Comparative Example 1, exhibited no change to the average strain to failure when compared to Example 1. Additionally, by decreasing the Mg content to 4.1 wt. % as provided in Example 3, the average strain to failure value increased to 0.176. Further decreasing the Mg content while increasing the Mn content as provided in Example 4 decreased the average strain to failure to 0.137. Moreover, by maintaining the same levels of Mn and Mg as Example 4 and increasing the content of Fe and Si, Example 5 exhibited an average strain to failure of 0.161, which is similar to Comparative Example 1. The results shown here indicate that by changing the alloy composition to contain lower amounts of Mg and increasing the content of Fe, Si, and Mn can maintain the same physical properties as Comparative Example 1 while significantly increasing its recycled content.

Example 2

[0182] Sample aluminum alloys were tested to determine the properties of aluminum alloys described herein. Comparative Examples 2 and 3 and Examples 7 and 8 were prepared according to the methods described herein. Comparative Example 1 was prepared from a conventional AA3104 aluminum alloy, which is currently employed as can body stock, and Comparative Example 2 has a similar

[0183] As shown in FIG. 12, Examples 7 and 8 achieve a buckle strength at or above 600 kPa. The buckle strength was measured using an End Buckling Test Station (from Altek Company, Model 9009114). Examples 7 and 8 have a similar composition to Comparative Examples 7 and 8, but have a higher Mg content. Since Examples 7 and 8 have a similar composition to Comparative Example 7 (which is AA3104 aluminum alloy), these aluminum alloys can be produced from higher amounts of recycled UBC scrap that contain similar amounts of Fe, Si, Cu, and Mn. This eliminates the need for diluting the aluminum alloys of Examples 7 and 8 with primary aluminum or adding additional hardening elements. In an example, to produce an aluminum alloy composition from recycled UBC, at least 2 wt. % of Mg may be added to cast an alloy composition with similar levels of Fe, Si, Cu, and Mn level in UBC as seen in Examples 7 and 8 from Table 21. Additionally, by maintaining the aluminum alloy compositions that may be similar to AA3104 aluminum alloy, the re-melting process may be simplified to reduce process changes during fabrication. The aluminum alloy compositions of Examples 7 and 8 have a similar Fe, Si, Cu, and Mn to conventional aluminum alloys for can body stock and can end stock, which allows re-melting process much simpler and make it a complete loop without alloy change. Furthermore, the recycled content will be significantly higher due to much lower needs of primary aluminum or hardening elements. In contrast, conventional AA5182 aluminum alloy would require dilution with Fe, Si, Cu, Mn, and addition of high amounts of Mg for casting from UBC.

Example 3

[0184] Sample aluminum alloys were investigated to determine the microstructure of aluminum alloys described herein when prepared according to specific homogenization conditions. Examples 9-12 are alloys described herein comprising various amounts of Mg and Cr. Table 23 provides the aluminum alloy composition for each of Examples 9-12.

TABLE 23

	Si (wt. %)	Fe (wt. %)	Cu (wt. %)	Mn (wt. %)	Mg (wt. %)	Cr (wt. %)	Bal	Description
Ex. 9	0.28	0.53	0.18	0.86	3.40	—	Al	
Ex. 10	0.28	0.53	0.18	0.86	2.90	—	Al	Mg reduction
Ex. 11	0.28	0.53	0.18	0.86	3.40	0.10	Al	Cr addition
Ex. 12	0.28	0.53	0.18	0.86	2.90	0.10	Al	Mg reduction and Cr addition

[0185] Standard thermodynamic calculations were performed for Examples 9-12 using Thermo-Calc Software (trade name, supplied by Thermo-Calc Software AB) to screen the effects of altering the aluminum alloy composition on the alpha phase transformation during homogenization. FIGS. 14A-14D show the equilibrium phase fraction diagrams of Examples 9-12. The x-axis represents the homogenization temperature ($^{\circ}$ C.) and the y-axis represents the volume fraction (mol) of different particles in the aluminum alloy. The green line represents the alpha phase particles (Al_6Mn) and the yellow line represents Mg_2Si . The navy blue line represents the liquidus temperature of the aluminum alloy (e.g. the temperature at which the aluminum alloy begins to melt). The gray shaded area designates the homogenization temperature window at which the alpha phase particles are stable in the aluminum alloy. FIGS. 14A-14D show the effects of the aluminum alloy composition on the homogenization temperature window for producing alpha phase particles. FIG. 14A shows that Example 9 has a very narrow homogenization temperature window for producing stable alpha phase particles that ranges from about 550° C. to about 600° C. However, a significant portion of the homogenization temperature window overlaps with the liquidus temperature at which the aluminum alloy melts, which is not desirable. This substantially limits the homogenization temperature range for producing alpha phase particles. FIG. 14A demonstrates that aluminum alloys produced from a high amount of recycled aluminum alloy materials have a narrow homogenization temperature window to produce alpha phase particles thereby limiting the amount of alpha phase in the aluminum alloy.

[0186] FIGS. 14B-D demonstrate that the stability of the alpha phase can be enhanced by modifying alloying elements as provided in Examples 9-12. Specifically, FIG. 14B shows the effect of Mg reduction on the homogenization temperature window compared to Example 9, FIG. 14C shows the effect of Cr addition on the homogenization temperature window compared to Example 9, and FIG. 14D shows the combined effect of Mg reduction and Cr addition on the homogenization temperature window compared to

Example 9. Each of Examples 10-12 have much larger homogenization temperature window for producing alpha phase particles. This provides higher amounts for phase transformation from large constituents to the alpha phase. Conducting homogenization at the temperature within the alpha stable region can result in the phase transformation from $\text{Alx}(\text{Fe},\text{Mn})$ to the alpha phase. As shown in FIGS. 14A-D, it was surprisingly found that the particle size for aluminum alloys produced from high amounts of recycled aluminum materials can be controlled by modifying alloying elements and homogenizing at temperatures that promote transformation of large constituents to alpha phase particles. [0187] Additionally, FIG. 14B shows that the homogenization temperature range is wider for to produce stable alpha phase particles by reducing Mg. Similarly, FIG. 14C shows that the homogenization temperature range for aluminum alloys including Cr is even larger for producing stable alpha phase particles. FIG. 14D shows that the homogenization temperature range for aluminum alloys including Cr and less Mg had the largest homogenization window for producing stable alpha phase particles. As discussed herein, alpha phase particles are beneficial because they can be broken down into smaller particles since alpha phase particles include eutectic formations that break down during hot rolling. Even though the volume fraction of the particles is increased, the particle size is substantially smaller. In contrast, $\text{Al}(\text{Fe},\text{Mn})$ is very bulky and hard particle that is difficult to breakdown. In some embodiments, the homogenization temperature range from 450° C. to 570° C. (e.g., from 500° C. to 560° C. or 525° C. to 550° C.) produces the highest concentration of stable alpha phase particles.

Example 4

[0188] Sample aluminum alloys were tested to investigate the amounts and sizes of particles in the microstructure of aluminum alloys described herein when prepared according to the methods described herein. Examples 13-16 are alloys include various amounts of Si, Cu, Mg, and Cr. Table 24 provides the aluminum alloy composition for each Examples 13-16.

TABLE 24

	Si (wt. %)	Fe (wt. %)	Cu (wt. %)	Mn (wt. %)	Mg (wt. %)	Cr (wt. %)	Bal	Description
Ex. 13	0.21	0.53	0.18	0.86	3.40	0	Al	
Ex. 14	0.28	0.53	0.18	0.86	3.40	0	Al	Si addition
Ex. 15	0.28	0.53	0.25	0.86	3.40	0	Al	Cu addition
Ex. 16	0.28	0.53	0.18	0.86	2.90	0.10	Al	Mg reduction and Cr addition

[0189] FIGS. 15A and 15B show the effects of the aluminum alloy composition on the particle area (%) and the average particle size in the microstructure for Examples 13-16. As shown in FIG. 15A, Example 13 had the highest amount of Alx(Fe,Mn) particles, which is a large constituent that is difficult to breakdown in rolling processes, and the least amount of alpha phase particles. FIG. 15B also shows that Example 13 had the largest average particle size of about 1.50 sq. microns. Example 14, which includes higher amounts of Si than Example 13, had less than 50% of the of Alx(Fe,Mn) particles compared Example 13. Additionally, Example 13 had an average particle size of about 1.45 sq. microns. The data demonstrates that Si addition to the aluminum alloy composition can result in more alpha phase transformation and smaller average particle sizes in the aluminum alloy microstructure.

[0190] Additionally, Example 15 shows the effects of Cu addition and Example 16 shows the effects of Mg reduction and Cr addition on the particle area (%) and the average particle size in the aluminum alloy microstructure. In each of Examples 15 and 16, the particle area (%) of alpha phase particles is greater than 1.8 wt. %. Both Examples 15 and 16 had little to no Alx(Fe,Mn) particles. The ratio of alpha phase particles to Alx(Fe,Mn) particles for Examples 15 and 16 was higher than Examples 13 and 14, with Example 16 having almost no Alx(Fe,Mn) particles in the aluminum alloy microstructure. Examples 15 and 16 also had particles that were less than 1.30 sq. microns. The examples demonstrate that the addition of Si, Cu, Cr, and the reduction of Mg, can beneficially transform large constituents into alpha phase particles during homogenization. Specifically, Example 16, which had lower Mg and additional Cr, resulted in the higher particle area % of alpha phase particles and had the smallest average particle size.

[0191] All patents, publications, and abstracts cited above are incorporated herein by reference in their entireties. Various embodiments of the invention have been described in fulfillment of the various objectives of the invention. It should be recognized that these embodiments are merely illustrative of the principles of the present invention. Numerous modifications and adaptations thereof will be readily apparent to those skilled in the art without departing from the spirit and scope of the present invention as defined in the following claims.

1. An aluminum alloy comprising 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

2. The aluminum alloy of claim 1, comprising 0.10-0.30 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-1.0 wt. % Mn, 2.2-5.0 wt. % Mg, up to 0.15 wt. % Cr, up to 0.30 wt. % Zn, up to 0.15 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

3. The aluminum alloy of claim 1, comprising 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.10 wt. % Cr, up to 0.25 wt. % Zn, up to 0.10 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

4. The aluminum alloy of claim 1, comprising 0.20-0.35 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.05 wt. % Cr, up to 0.25 wt. % Zn, up to 0.05 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

5. The aluminum alloy of claim 1, comprising about 0.20-0.35 wt. % Si, 0.40-0.60 wt. % Fe, 0.15-0.25 wt. % Cu, 0.60-1.2 wt. % Mn, 2.0-4.0 wt. % Mg, up to 0.03 wt. % Cr, up to 0.20 wt. % Zn, up to 0.03 wt. % Ti, up to 0.15 wt. % impurities, and Al.

6. The aluminum alloy of claim 1, wherein a ratio of Mg:Cu is from 10:1 to 80:1 and wherein a ratio of Mn:Cu is from 2:1 to 15:1.

7. The aluminum alloy of claim 1, wherein a ratio of Mg:Cu is from 15:1 to 70:1 and wherein a ratio of Mn:Cu is from 3:1 to 12:1.

8. The aluminum alloy of claim 1, wherein a combined content of Fe and Si is greater than 0.40 wt. %, and wherein a combined content of Mg, Mn, and Cu is from 3.5 wt. % to 5.0 wt. %.

9. (canceled)

10. The aluminum alloy of claim 1, wherein the aluminum alloy comprises at least 40 wt. % of recycled scrap, and wherein the aluminum alloy comprises less than 30 wt. % of primary aluminum.

11. (canceled)

12. The aluminum alloy of claim 1, wherein the aluminum alloy has a propagation energy of at least 30.0 KJ/m² as measured by ASTM B871-1 (2021).

13. The aluminum alloy of claim 1, wherein the aluminum alloy has a yield strength of at least 340 MPa, wherein the aluminum alloy has an ultimate tensile strength of at least 380 MPa, and wherein the aluminum alloy has a total elongation of at least 4%.

14. (canceled)

15. (canceled)

16. A can end stock comprising the aluminum alloy of claim 1.

17. A method of producing an aluminum alloy comprising:

casting an aluminum alloy to form a cast product, wherein the aluminum alloy comprises 0.10-0.35 wt. % Si, 0.20-0.60 wt. % Fe, 0.05-0.25 wt. % Cu, 0.25-1.20 wt. % Mn, 2.0-5.0 wt. % Mg, up to 0.20 wt. % Cr, up to 0.30 wt. % Zn, up to 0.20 wt. % Ti, up to 0.15 wt. % of impurities, and Al;

homogenizing the cast product;

hot rolling the cast product to produce a hot rolled product;

cold rolling the hot rolled product to produce a final gauge rolled product; and

optionally annealing the final gauge rolled product.

18. The method of claim 17, further comprising lacquering and curing the final gauge rolled product.

19. The method of claim 17, wherein the homogenization step comprises a first homogenization step and a second homogenization step.

20. The method of claim 19, wherein the first homogenization step comprises soaking the cast product at a temperature from 375° C. to 450° C. for 0.5 hours to 5 hours.

21. The method of claim 19, wherein the second homogenization step comprises soaking the cast product at a temperature from 450° C. to 550° C. for 0.01 hours to 5 hours.

22. The method of claim 17, wherein the aluminum alloy comprises 0.10-0.25 wt. % Si, 0.20-0.50 wt. % Fe, 0.05-0.25 wt. % Cu, 0.30-0.90 wt. % Mn, 2.5-5.0 wt. % Mg, up to 0.10 wt. % Cr, up to 0.25 wt. % Zn, up to 0.10 wt. % Ti, up to 0.15 wt. % of impurities, and Al.

23. The method of claim **17**, wherein a ratio of Mg:Cu is from 15:1 to 70:1 and wherein a ratio of Mn:Cu is from 3:1 to 12:1.

24. A metal product, wherein the metal product is prepared by a method comprising claim **17**, wherein the metal product is can end stock.

25. (canceled)

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